

Differential roles of Nav1.2 and Nav1.6 in neocortical pyramidal cell excitability

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This manuscript presents a clever and powerful approach to examining differential roles of Nav1.2 and Nav1.6 channels in excitability of neocortical pyramidal neurons, by engineering mice in which a sulfonamide inhibitor of both channels has reduced affinity for one or the other channels. Overall, the results in the manuscript are **compelling** and give **important** information about differential roles of Nav1.6 and Nav1.2 channels. Activity-dependent inactivation of Nav1.6 was also found to attenuate seizure-like activity in cells, demonstrating the promise of activity-dependent Nav1.6-specific pharmacotherapy for epilepsy.

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Abstract

Mature neocortical pyramidal cells functionally express two sodium channel (Nav) isoforms: Nav1.2 and Nav1.6. These isoforms are differentially localized to pyramidal cell compartments, and as such are thought to contribute to different aspects of neuronal excitability. But determining their precise roles in pyramidal cell excitability has been hampered by a lack of tools that allow for selective, acute block of each isoform individually. Here, we leveraged aryl sulfonamide-based molecule (ASC) inhibitors of Nav channels that exhibit state-dependent block of both Nav1.2 and Nav1.6, along with knock-in mice with changes in Nav1.2 or Nav1.6 structure that prevents ASC binding. This allowed for acute, potent, and reversible block of individual isoforms that permitted dissection of the unique contributions of Nav1.2 and Nav1.6 in pyramidal cell excitability. Remarkably, block of each isoform had contrasting—and in some situations, opposing—effects on neuronal action potential output, with Nav1.6 block decreasing and Nav1.2 block increasing output. Thus, Nav isoforms have unique roles in regulating different aspects of pyramidal cell excitability, and

our work may help guide development of therapeutics designed to temper hyperexcitability through selective Na_v isoform blockade.

Introduction

Voltage-gated sodium channels (Na_v) are critical for all aspects of neuronal excitability, from action potential (AP) initiation to axonal propagation, transmitter release, and dendritic excitability^{1–6}. In mature neocortical pyramidal cells, electrogenesis and subsequent propagation of APs is systematically regulated with Na_v recruitment first initiated in the axon initial segment (AIS), with forward propagation along the axon and backpropagation into soma and dendrites^{2,3,7}. This coordinated process is supported by membrane expression of the Na_v isoforms Na_v1.6 and Na_v1.2. Current models, based on immunostaining of channels and empirical measurements of excitability, suggest that Na_v1.2 and Na_v1.6 are differentially expressed across neurites. In the AIS, Na_v1.6 predominates, with highest membrane density in the AP initiation region of the AIS most distal to the soma. Na_v1.2, by contrast, has higher relative membrane density in the proximal AIS^{2,8–12}. Along the axon, Na_v1.6 appears enriched at nodes of Ranvier and terminals^{13,14}. Somatodendritic densities are far lower, and current models suggest that the somatic and perisomatic membrane expresses equal levels of Na_v1.6 and Na_v1.2, whereas dendritic regions more distal to the soma are enriched exclusively with Na_v1.2^{4–6,15}. Thus, each channel isoform likely has a unique role in AP initiation and propagation simply based on their differential distribution in neuronal compartments.

To date, efforts to understand the unique contributions of different Na_v isoforms has relied largely on experiments in which channel genetic expression has been manipulated, either through constitutive or conditional knockout approaches. While these approaches have strong merit, interpretation is complicated by compensatory changes in other Na_v isoforms or other ion channel classes^{4,5,16,17}. Pharmacological approaches, which are both acute and reversible, would be preferred, but identifying highly selective compounds that target particular Na_v isoforms is difficult, as *ScnXa* gene family isoforms have high degrees of amino acid sequence homology. Some compounds exhibit high potency, but differences in the half-maximal inhibitory concentration (IC₅₀) for individual isoforms can be limited^{18–21}. This is especially true for separation of Na_v1.2 from Na_v1.6, as these two channels have especially high sequence similarity^{22,23}.

Aryl sulfonamide compounds (ASCs) constitute a unique class of Na_v inhibitors that potently bind activated Na_v isoforms, resulting in a stabilization of the inactivated state^{24–28}. The binding pocket is shielded from ASCs when channels are closed due to the positioning of the S4 voltage sensing domain, but is bound rapidly by ASCs when exposed. Unbinding is promoted by strong hyperpolarization, but occurs much more slowly at physiological voltages (e.g., tau of 10²–10³ sec at –80 and –40 mV, respectively)²⁴. In spiking neurons, this essentially imparts use-dependence to ASC block²⁴, as the probability of channel block will increase in proportion with AP rates.

ASCs exhibit high potency for a subset of Na_v isoforms: Na_v1.2, Na_v1.6 and Na_v1.7^{24–27}. These isoforms share sequence homology at the binding pocket, each containing a tyrosine-tryptophan (YW) motif that helps stabilize ASC binding. By contrast, Na_v1.1 and Na_v1.3 harbor a serine-arginine (SR) sequence at the same site. Mutagenesis of Na_v1.7 from YW to SR results in a 145-fold decrease in ASC binding without affecting channel biophysical properties²⁴. Here, we engaged a similar strategy for Na_v1.2 and Na_v1.6, using knock-in mice with YW→SR substitutions in either or both channels. This allows for selective, potent ASC-mediated block of YW-containing channels while preserving the function of SR-containing channels. With these tools, we dissected the individual roles of Na_v1.2 and Na_v1.6 in neocortical pyramidal cell excitability. We found that Na_v1.2 and Na_v1.6 have unique—and at times conflicting—effects on overall AP excitability. Moreover, results suggest that ASCs can function as ‘on-demand’ pharmacological inhibitors that

help normalize cellular excitability in seizure-like conditions. Together, this work highlights the importance of distinct Na_V isoforms localized to specific cellular domains and their respective contribution to AP properties to help facilitate activity across pyramidal neurons.

Results

Aryl sulfonamides selectively bind and inhibit Na_V isoforms containing the YW motif

ASCs exhibit high affinity for an extracellular region within voltage sensing domain IV (VSD IV) of select Na_V channels with a conserved tyrosine-tryptophan (YW) motif²⁴ (Fig. 1A, B). This YW motif found on Na_V1.2, Na_V1.6 and Na_V1.7 allows for ASC stabilization following channel inactivation (Fig. 1B). By contrast, an SR motif present in Na_V1.1 and Na 1.3 limits binding appreciably²⁴. Thus, we hypothesized that converting either Na_V1.2 or Na_V1.6 channels from those that contain the YW motif to those that contain an SR motif would alter ASC binding to those channels significantly.

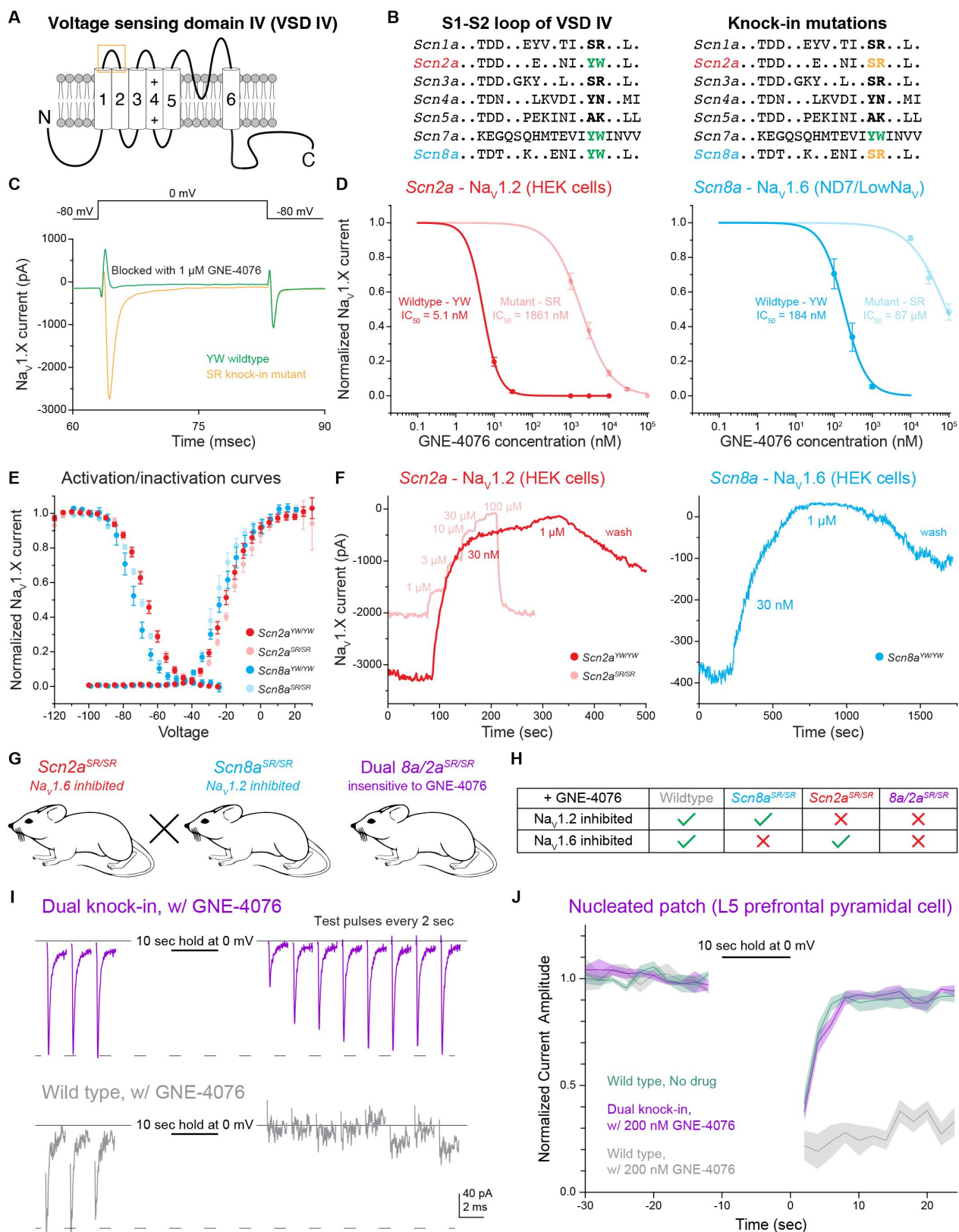


Figure 1

The YW motif on Na_v1.2 and Na_v1.6 increases activity-dependent GNE-4076 potency and subsequent channel inhibition

(A) Schematic depicting the fourth voltage sensing domain (VSD-IV) of Na_v isoforms. The six transmembrane spanning regions have high sequence homology amongst different Na_v isoforms, while linker regions display more sequence divergence. Orange box highlights extracellular S1-S2 loop where ASCs are stabilized by a tyrosine-tryptophan (YW) motif.

(B) Amino acid sequence within the S1-S2 loop of various Na_v isoforms. *Scn2a* (Na_v1.2) and *Scn8a* (Na_v1.6) are the predominant channels expressed in mature, prefrontal pyramidal cells. Both isoforms share a conserved YW sequence that increases ASC potency. Knock-in mutations of *Scn2a* and *Scn8a* were generated by substituting the YW motif with a serine-arginine (SR) sequence present in *Scn1a* and *Scn3a*.

(C) Example Na_v current traces (pA) of cells expressing either YW wildtype channels or SR knock-in mutant chimeras in the presence of 1 μM GNE-4076. To activate exogenously expressed Na_v channels, cells were held at -80 mV and stepped to 0 mV for 20 ms. GNE-4076 onboarding was performed by holding cells at 0 mV for 10 sec.

(D) Dose response curves for exogenously expressed *Scn2a* (HEK cells) or *Scn8a* (ND7/LoNa_v) in immortalized cell lines. IC₅₀ was measured for both YW wildtype channels and SR knock-in mutant chimeras. YW->SR knock-in mutations reduced GNE-4076 potency by about 400-to 500-fold relative to wildtype channels. Circles represent normalized mean Na_v current amplitude ± SEM.

(E) Activation and steady-state inactivation curves for both YW wildtype channels and SR knock-in mutant chimeras (*Scn2a*^{YW/YW}, n=6; *Scn2a*^{SR/SR}, n=7; *Scn8a*^{YW/YW}, n=8; *Scn8a*^{SR/SR}, n=6). *Scn2a* or *Scn8a* YW->SR mutations alter efficacy of GNE-4076 while having minor effects on biophysical properties of either isoform. Circles represent mean normalized Na_v current amplitude ± SEM. Unpaired t-test with Welch's correction. No significance detected between wildtype and mutant channels for both V1/2 of activation or inactivation.

(F) Example current amplitude response graphs for Na_v1.2 (red) and Na_v1.6 (blue) expressed in HEK cells. Cells were perfused with increasing concentrations of GNE-4076 throughout the recording. Individual current response recordings from HEK cells expressing *Scn2a* were robust (3.2 nA), and recordings were reproducible for both YW wildtype channels (red) and SR knock-in mutant chimeras (transparent red). Current responses from cells expressing *Scn8a* were variable with only a few cells exhibiting channel conductance (400 pA). In select cells expressing *Scn8a*, current amplitude (blue) also decreases substantially with 30 nM GNE-4076 and completely with 1 μM.

(G) Transgenic mouse lines generated with the YW->SR knock-in mutation present on both *ScnXa* alleles. *Scn2a*^{SR/SR} mice were crossed with *Scn8a*^{SR/SR} mice to generate a dual *8a/2a*^{SR/SR} knock-in mouse.

(H) Overview of the various transgenic (or wildtype) mouse lines used throughout this study. Application of 200 nM GNE-4076 selectively inhibits Na_v isoforms only containing the YW motif.

(I) Nucleated patch experiments from prefrontal pyramidal cells performed in wildtype or dual *8a/2a*^{SR/SR} knock-in cells in the presence of 200 nM GNE-4076. Baseline conductance was measured by depolarizing cells from -80 mV to -12 mV every 2 sec for 10 pulses. Na_v channels were inactivated by holding nucleated patch at -12 mV for 10 sec. Test pulses were again acquired during recovery similar to baseline pulses.

(J) Summary graph of normalized current amplitude from nucleated patch experiments in (I). Baseline and recovery test pulses were acquired for at least 20 sec before and 25 sec after channel inactivation step. Solid line represents normalized mean Na_v current amplitude ± SEM. Graph also include wildtype, no drug control nucleated patch experiments (wildtype no drug, n=4; wildtype + GNE-4076, n=4; *8a/2a*^{SR/SR} + GNE-4076, n=4).

To test for the effects of this motif substitution, we first examined currents generated by wild type and YW->SR mutated channels expressed in immortalized cell lines (**Fig. 1C**). HEK cells were primarily used for most experiments. But due to known low transfection efficiency of Na_v1.6 in HEK cells, a subset of experiments were performed using an ND7/23 cell line engineered to lack most native Na_v conductance (ND7/LoNa)³⁵. Both Na_v1.2 and Na_v1.6 wildtype channels expressed in cell lines were inhibited markedly by 1 μM GNE-4076 (**Fig. 1C**). By contrast, SR knock-in mutants continued to flux sodium in the presence of 1 μM GNE-4076. Dose response curves revealed an IC₅₀ of 5.1 nM for wildtype Na_v1.2 that increases 365-fold to 1861 nM for mutant SR channels (**Fig. 1D**). For Na_v1.6 channels, GNE-4076 IC₅₀ increased 475-fold from 184 nM to 87 μM with the SR mutant (**Fig. 1D**). To assess whether SR mutations affect channel gating properties in the absence of GNE-4076 binding, steady state activation and inactivation curves were assessed for wildtype or mutant channels (**Fig. 1E**). Consistent with prior work studying a similar mutation in Na_v1.7²⁴, the YW to SR mutation had no effect on slope factor for both voltage-dependent activation or inactivation for either Na_v1.2 or Na_v1.6 (Table 2); however, both voltage-dependent activation and inactivation hyperpolarized by approximately 2 mV for Na_v1.6, but not Na_v1.2.

While GNE-4076 binds potently to either Na_v1.2 (**Fig. 1D, F**) or Na_v1.7 expressed in HEK cells²⁸, its affinity was lower for Na_v1.6 expressed in ND7/LoNa cells. To test if this was due to reductions in affinity imposed by the ND7/23 cell line, we examined current in the few HEK cells in which Na_v1.6 current could be measured (**Fig. 1F**). While peak currents were small (400 pA vs 3.2 nA for Na_v1.2 transfected using identical protocols), we found that 30 nM GNE-4076 exhibited strong block of Na_v1.6 (**Fig. 1F**), but due to limitations in expression efficiency in HEK cells, we were unable to collect full dose-response curves to determine IC₅₀.

Results in heterologous expression systems indicate that YW->SR substitution reduces GNE-4076 binding affinity markedly. Given these results, we constructed transgenic mice containing the YW->SR mutation. Two distinct lines, termed *Scn8a*^{SR/SR} and *Scn2a*^{SR/SR}, were generated by mutating both alleles of each sodium channel gene (**Fig. 1G**). SR/SR mutations were validated by PCR (see Methods) and mice were bred to be homozygous for the mutations. Comparisons were then made between wild type mice, mice with YW->SR mutations into Na_v1.2 or Na_v1.6 alone (*2a*^{SR/SR} or *8a*^{SR/SR}, respectively), or YW->SR mutations in both Na_v1.6 and Na_v1.2 (dual *8a/2a*^{SR/SR}). In principle, one could determine a concentration of GNE-4076 in which native channels are blocked potently, sparing SR variant channels (**Fig. 1H**). Thus, these mice should enable selective, acute manipulations of Na_v1.2 and Na_v1.6-dependent aspects of neuronal excitability using a chemical genetics approach.

Given the disparate binding results in heterologous cell lines, we first tested the effects of a dose that should potently block WT channels but not affect SR variant channels to verify that native channels behave similarly to those expressed in HEK cells. Neocortical pyramidal cell somata are thought to express similar levels of Na_v1.2 and Na_v1.6 on their membranes^{4,5,8}. Thus, excised patches from this region can provide insight into block of both isoforms. Acute coronal slices containing mPFC were prepared from WT or *8a/2a*^{SR/SR} mice and nucleated patches were excised from layer 5 pyramidal cells. After establishing a stable baseline of Na_v-mediated current evoked from a holding voltage of -80 mV, a value comparable to the resting membrane potential studied in later current-clamp experiments, cells were pulsed to -12 mV for 10 sec to achieve near-complete block of Na_v isoforms, then returned to -80 mV (**Fig. 1I**). In untreated WT conditions, currents recovered to near-baseline levels within 5 sec (**Fig. 1J**). A similar quick recovery was observed in cells from *8a/2a*^{SR/SR} mice in the presence of 200 nM GNE-4076. By contrast, WT neurons exposed to the same dosing were inhibited markedly, with no appreciable recovery within 25 sec (**Fig. 1I, J**). Taken together, these results show that the YW to SR mutation reduces ASC binding and inhibition of Na_v chimeras both in immortalized cell lines and pyramidal neurons.

We also asked whether neuronal AP properties were affected by these mutations in $8a/2a^{SR/SR}$ mice by assessing both threshold and peak dV/dt in the absence of GNE-4076 (Fig. S1). Compared to wildtype cells, there were no detectable changes to peak dV/dt in $8a/2a^{SR/SR}$ neurons (Fig. S2B). However, AP threshold was hyperpolarized in $8a/2a^{SR/SR}$ neurons, likely due to small changes in voltage-dependence of activation observed for heterologously expressed $Na_v1.6$ (Fig. S2A, Table 2). Application with GNE-4076 in either wildtype or $8a/2a^{SR/SR}$ neurons had no further effect on threshold or peak dV/dt compared to controls recording in the absence of GNE-4076, suggesting minimal drug binding occurs without marked depolarization or spiking activity (Fig. S1, Table 3).

Differential roles for $Na_v1.6$ and $Na_v1.2$ in AP initiation and somatic excitability

$Na_v1.2$ and $Na_v1.6$ are differentially distributed in neuronal arbors and are thought to contribute to different aspects of AP initiation and propagation. In mature neocortical pyramidal cells, $Na_v1.6$ is enriched in the distal AIS, the site where APs initiate^{4,8,16,44}. Following initiation, APs forward propagate along the axon via $Na_v1.6$ -enriched nodes of Ranvier, but also backpropagate into the soma through a region enriched with a mix of $Na_v1.2$ and $Na_v1.6$ ⁸. Several studies have sought to identify specific roles for each isoform by conditional deletion of either isoform^{4,16}. Unfortunately, in such conditions, the residual isoform compensates for loss to some degree, making interpretation of individual isoform roles difficult. We therefore leveraged ASC-based block to study acute, differential inhibition of Na_v isoforms to better understand the individual roles of $Na_v1.2$ and $Na_v1.6$ in AP excitability.

Changes to AP waveform were visualized with phase plane plots, which plot membrane voltage vs. membrane voltage velocity (Fig. 2B⁷ and S3A). Within phase plots, spike threshold appears as a sudden deviation from rest and is defined as the voltage at which voltage velocity (dV/dt) first exceeds 15 V/s (Fig. S3A). Following spike threshold, a neuron's voltage transits through two components of Na_v recruitment, first in the AIS and then in the soma^{7,45}. These result in characteristic “humps” in the depolarizing aspect of the phase plot^{2,11}. Thus, these different components of the phase plot can aid one's understanding of effects on specific components of Na_v -mediated excitability.

To provide *a priori* predictions of potential ASC-based effects on excitability, we constructed a compartmental model in which each channel's density could be modulated. Channels were distributed based on predictions from empirical anatomical and physiological studies^{4,6,8,16,46}. $Na_v1.6$ was enriched in the distal AIS, $Na_v1.2$ was enriched in the proximal AIS, both channels were expressed at equal levels in the soma and the first 20 microns of dendrite closest to the soma, and $Na_v1.2$ was expressed exclusively in all other dendrites (Fig. 2A⁷). Within this model, each channel's density was modulated in 10% increments, from 100 to 0% (Fig. 2B⁷).

In WT conditions (100% density of both channels), models generated APs with a threshold and AP kinetics comparable to empirical baseline observations (Fig. 2B, C⁷). Within this model, reducing $Na_v1.2$ or $Na_v1.6$ density had clear, dissociable effects. Progressive reduction of $Na_v1.6$ produced a progressive depolarization of AP threshold and a corresponding decrement in dV/dt throughout the entire rising phase of the AP. $Na_v1.2$ reduction, by contrast, had no effect on AP threshold or the initial velocity of the AP initiated in the distal AIS. Instead, components of the AP related to backpropagation and recruitment of somatic Na_v channels were impaired only, with a decrement in peak dV/dt and peak AP membrane potential. To validate the accuracy of the parameters used in our model, we also performed sensitivity analysis (Fig. S2). Here, we adjusted the crossover point of $Na_v1.2$ and $Na_v1.6$ within the AIS as well as varied the density ratio of both channels (Fig. S2A, B). Increasing $Na_v1.6$ ratio to 100% consistently hyperpolarized AP threshold across all AIS crossover positions (Fig. S2B, C). However, dV/dt was highly variable and random when either the AIS crossover point is shifted or Na_v ratios are altered.

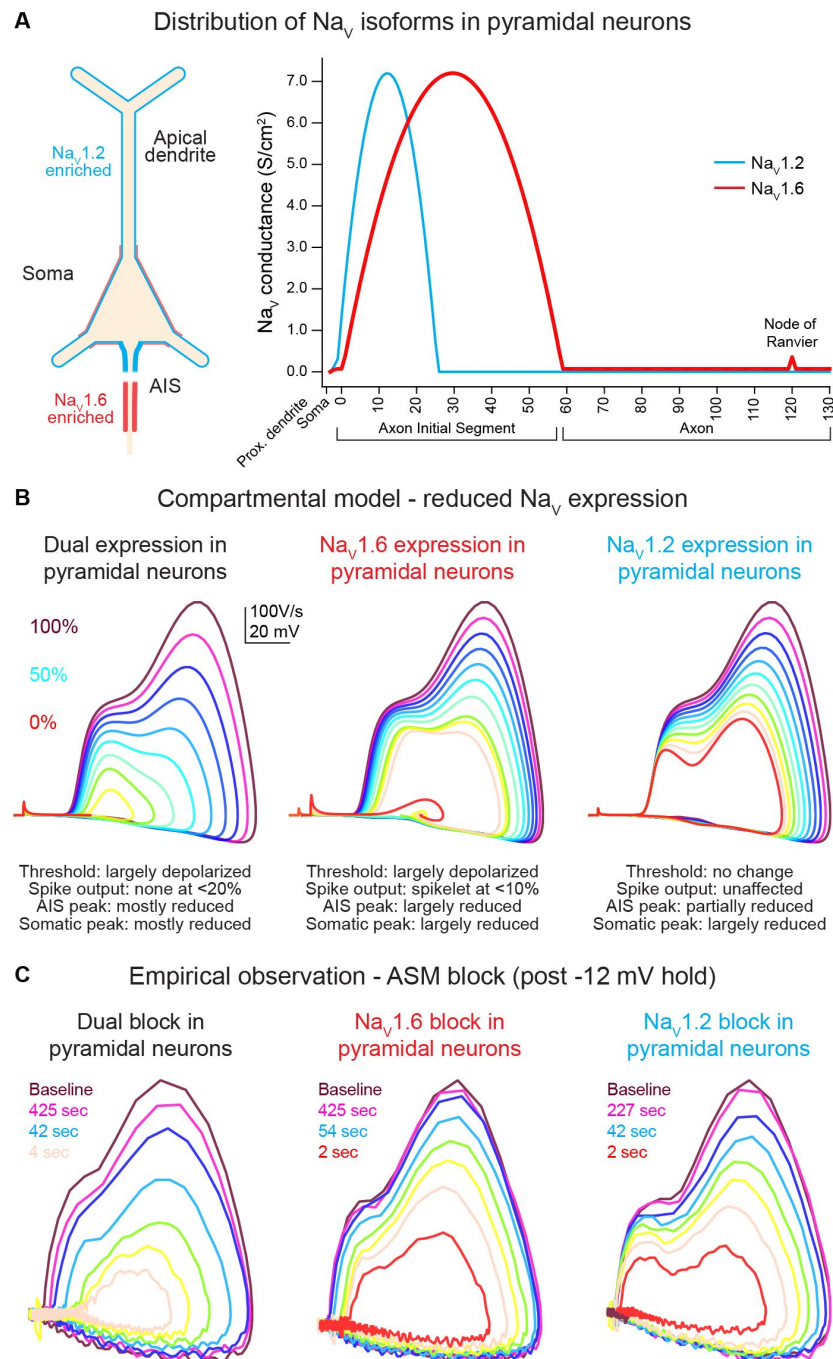


Figure 2

Global *Scn2a*, *Scn8a* or dual loss in compartmental models distinctly impacts key AP properties

(A) $\text{Na}_v1.6$ and $\text{Na}_v1.2$ are equally expressed in soma and proximal dendrites. Expression pattern is more distinct in other regions with $\text{Na}_v1.6$ enriched in the distal AIS, axon and nodes of Ranvier, whereas $\text{Na}_v1.2$ is found exclusively in the proximal AIS and distal dendrites.

(B) Compartmental model representing changes to phase plots when Na_v isoform expression is reduced from 100% (warmer colors) to 0% (in 10% increments) based on known localization across distinct neuronal localities. Lower $\text{Na}_v1.6$ expression depolarized spike threshold and decreased both AIS and somatic AP velocity (dV/dt). Reduced $\text{Na}_v1.2$ expression largely impacts backpropagation and somatic AP velocity.

(C) Empirical observations of phase plot following near complete channel block with ASCs. Darker trace represents phase plot taken at baseline prior to -12 mV hold for 30 sec (see **Fig. 3A**). Colored traces represent recovery phase plots at different times post -12 mV hold for 30 sec with warmer colors depicting more time lapsed and increased channel recovery.

While these models provide clues to the differential roles of $\text{Na}_V1.2$ and $\text{Na}_V1.6$, we needed to design an experimental strategy that increased the overall degree of Na_V blockade using ASCs in mice where either $\text{Na}_V1.6$ or $\text{Na}_V1.2$ was mutated to be insensitive to 200 nM GNE-4076 binding (Fig. 3C, 1G). Since it is nearly impossible to achieve full block with ASCs under physiological firing, we developed a hybrid current- and voltage-clamp experiment to study effects of more complete isoform blockade (Fig. 3A). In this protocol, neurons were held to -80 mV in current clamp with constant bias current (if necessary) and baseline APs were elicited with brief somatic current injection (300 ms, amplitude adjusted to evoke ~ 4 -5 APs). We then promoted Na_V activation and inactivation by voltage-clamping neurons to -12 mV for 30 sec (Fig. 3A). Following voltage-clamp, neurons were returned to current-clamp with the same bias current as used in baseline conditions. We then evoked APs, first at an interstimulus interval of 2 sec, then at successively increasing intervals of 5 to 60 sec, allowing for somewhat consistent sampling on a log-base timescale that aligns well with both channel recovery from inactivation and recovery from GNE-4076 block (Fig. 3D, F). For clarity and simplicity, we will describe studies based on which channel is inhibited rather than which channel was rendered insensitive to GNE-4076. For example, a study in the $\text{Scn2a}^{\text{SR/SR}}/\text{Scn8a}^{+/+}$ animal is a case where $\text{Na}_V1.6$ can be blocked.

Empirical data at various timepoints post -12 mV hold reveals that recovery from ASC-dependent block increases channel availability from approximate 0-10% to about 90-95% of the baseline spike when more time has elapsed (Fig. 2C). This recovery in available channels mimics changes observed to phase plots in compartmental models based on overall channel expression (Fig. 2B).

We then focused on our analysis on recovery of threshed or peak dV/dt across all genotypes (Fig. 3). In untreated WT cells, AP threshold following voltage-clamp recovered to baseline values immediately (Fig. 3C, D). By contrast, peak somatic dV/dt recovered more slowly (Fig. 3C, F), likely reflecting recovery from slow inactivation of channels in the perisomatic region⁴⁷⁻⁴⁹. Peak dV/dt recovered to within $87.12 \pm 0.86\%$ by 12-20 sec of voltage-clamp offset (Fig. 3F, G). Subsequent recovery of the residual peak dV/dt was slower, taking another 15-30 sec to recover the next 5% of peak dV/dt (Fig. 3F). Identical results were observed in $8a/2a^{\text{SR/SR}}$ cells treated with 200 nM GNE-4076, suggesting that any effect observed in single knock-in recordings will be due to block of GNE-4076 sensitive channels (Fig. 3B-G). We therefore focused on the 12-20 sec after voltage-clamp offset for subsequent analysis, as it is a period in which most channel-intrinsic recovery has occurred, but also a period in which we would still expect significant block from GNE-4076.

When individual channel isoforms were blocked more completely with voltage steps to -12 mV, dramatic changes in AP threshold and peak dV/dt were observed. When $\text{Na}_V1.6$ was blocked, threshold depolarized by 5.8 ± 0.7 mV (Fig. 3E). When $\text{Na}_V1.2$ was blocked instead, threshold was unaffected (ΔV_m : -0.1 ± 0.2). Thus, AP threshold and AP initiation appears to be initiated in an $\text{Na}_V1.6$ -rich region in control conditions; but when $\text{Na}_V1.6$ is inhibited, APs can occur at more depolarized potentials, likely mediated predominately by $\text{Na}_V1.2$.

To examine these relative contributions further, we analyzed the rising phase of APs more closely (Fig. S3A). As described above, the first component of the rising phase of the AP reflects recruitment of Na_V channels localized to the AIS (Fig. S3A, middle). Based on changes in voltage acceleration (second derivative) within this period, the AIS component can be further divided into initiation and AIS backpropagation components^{8,37} (Fig. S3A, right). These periods were divided based on a timepoint during the AIS phase where voltage acceleration first peaked (AIS inflection point) or afterwards decreased, creating a trough in an acceleration vs. time graph (AIS max; Fig. S3B). The dV/dt at the AIS inflection point was affected only by $\text{Na}_V1.6$ block (Fig. S3C), whereas the dV/dt at the trough was affected by block of either isoform (Fig. S3D, E). This reinforces the model where APs are initiated via the $\text{Na}_V1.6$ -enriched distal AIS, and where the

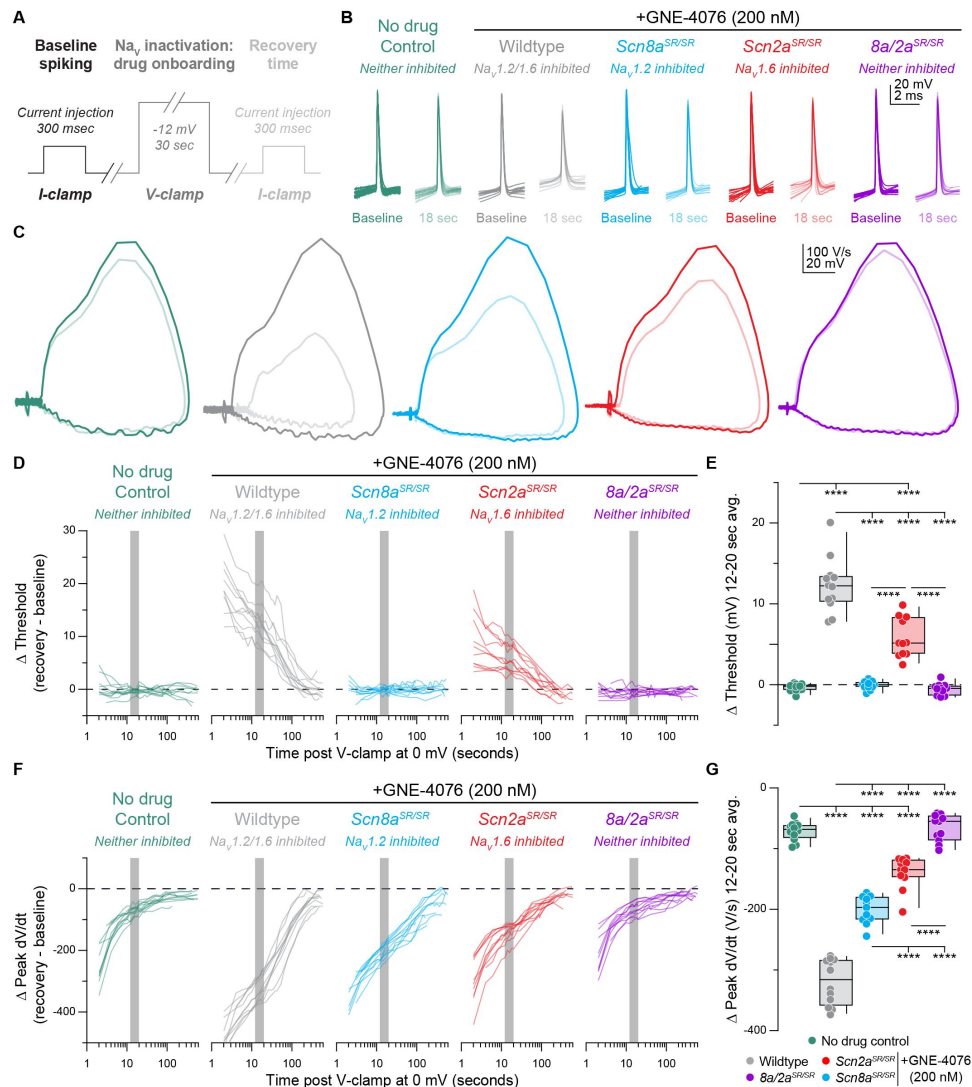


Figure 3

Recovery of AP firing properties is greatly diminished following dual $\text{Na}_V1.6$ and $\text{Na}_V1.2$ inhibition compared to selective block of individual channels

(A) Protocol used to characterize recovery of AP firing properties. Baseline spiking is determined by injecting current for 300 ms to elicit 5–6 APs. To promote Na_V inactivation and maximal GNE-4076 onboarding, neurons are held at -12 mV in voltage-clamp for 30 sec. Recovery of AP firing is evaluated by injecting same current stimulus defined during baseline spiking with an inter-stimulus interval starting at 2 sec, followed by 5, 15, 30 and 60 sec.

(B) Overlaid waveform of 1st AP at baseline or 18 sec post GNE-4076 onboarding for all conditions (wildtype no drug, $n=12$; wildtype + GNE-4076, $n=11$; $8a/2a^{\text{SR/SR}}$ + GNE-4076, $n=12$; $\text{Scn}2a^{\text{SR/SR}}$ + GNE-4076, $n=12$; $\text{Scn}8a^{\text{SR/SR}}$ + GNE-4076, $n=12$).

(C) Overlaid phase plane of AP traces at baseline (100% transparency) or 18 sec post GNE-4076 onboarding (20% transparency) for each condition in (B). Plots represent the AP velocity by taking the first derivative (dV/dt , y-axis) versus the membrane potential (mV, x-axis). Colors are matched to conditions represented in (B).

(D) Recovery of AP threshold (V_m) represented as a delta value for individual cells plotted against time post GNE-4076 onboarding (log-scale). For ΔV_m , baseline value is subtracted from individual timepoints throughout the recovery phase (Δ mV = recovery timepoint – baseline). Colors are matched to conditions represented in (B). Gray shaded bar represents recovery between 12–20 sec.

(E) Summary data for ΔV_m at 12–20 sec post GNE-4076 onboarding (time period represented as gray bar in (D)). Box plots show median and 90% tails. Circles represent individual cells. One-way ANOVA, Holm-Šidák multiple comparisons test.

**** $p < 0.0001$.

depolarization produced by these distally localized channels recruits $\text{Na}_V1.2$ in the more proximal AIS. Absolute values are also reported for threshold, peak dV/dt and AIS max for all recorded cells in recovery experiments (Fig. S4).

Acute block of $\text{Na}_V1.2$ increases pyramidal cell AP output

Previously, we showed that conditional knockout of *Scn2a* increased AP output⁴. We hypothesized that this was due to the lack of $\text{Na}_V1.2$ in dendrites. When they are absent from this compartment, the dendrite does not depolarize as effectively, leading to a corresponding decrease in the recruitment of dendrite-localized potassium channels (K_V). Consequently, neurons repolarize less between APs, making it easier to evoke the next AP in the $\text{Na}_V1.6$ -enriched AIS.

Though this hypothesis was supported by compartmental modeling demonstrating that acute block of $\text{Na}_V1.2$ mirrored empirical observations, we could not eliminate the possibility that some form of cellular compensation occurred in the weeks between conditional knockout induction and acute slice experiments⁵⁰. Therefore, we leveraged GNE-4076 to test the effects of acute $\text{Na}_V1.2$ block (Fig. 4A), eliminating the possibility of genetic or post-translational compensation. We further compared this manipulation to block of $\text{Na}_V1.6$ or both isoforms by examining overall AP output from protocols described in Fig. 3A.

Remarkably, acute $\text{Na}_V1.2$ block mirrored conditional knockout, and was the only condition in which AP output increased (Fig. 4B) and was associated with a depolarization in afterhyperpolarization voltage (Fig. 4C). In contrast, $\text{Na}_V1.6$ block decreased AP output, as did block of both channels (Fig. 4B). This indicates that blocking $\text{Na}_V1.2$ alone can increase AP output, independent of compensatory changes to other channels that may occur with genetic manipulations.

Activity-dependent effects of ASCs on Na_V channels and subsequent inhibition alters AP properties and firing rate

During physiological activity, non-selective pharmacological inactivated state^{24–27}. In contrast to data presented above where we observe near complete block immediately following prolonged depolarization (Fig. S4, absolute values), neurons experiencing physiological levels of activity likely never reach high percentages of channel blockade. Thus, in normal spiking neurons, ASCs would therefore be predicted to exhibit use-dependence, progressively blocking channels in proportion to a neuron's activity rate.

To test this concept, we generated prolonged periods of activity in current-clamp by injecting 300 pA into the somatic pipette for 10 sec (Fig. 5A). Drug-naïve WT neurons responded to somatic current injection with inhibition of Na_V and AP firing channels in neurons greatly reduces cellular excitability repetitive spiking at a rate of 14.7 ± 0.5 Hz ($n = 12$). Neurons fired at a steady state after the first second, with stable AP threshold and instantaneous AP^{51,52}. ASCs are unique in that they require prolonged bursts of activity or even neuronal hyperexcitability to stabilize more channels in the frequency (Fig. 5C). Spiking characteristics were identical in $8a/2a^{\text{SR/SR}}$ cells treated with GNE-4076, indicating that, at 200 nM, neurons are insensitive to GNE-4076 when the YW→SR motif substitution is present (Fig. 5C, D). However, in WT neurons treated with GNE-4076, threshold continued to depolarize after the first second, with corresponding decrements in instantaneous frequency and peak AP dV/dt throughout the stimulus (Fig. 5C, D). This demonstrates that GNE-4076 suppresses firing of neurons with a baseline firing rate of ~15 Hz (Table 4).

We then repeated experiments described above—where APs were generated with somatic depolarization over 10 sec (Fig. 6). In cases where $\text{Na}_V1.6$ alone could be inhibited, AP threshold depolarized in response to GNE-4076 as much as in WT cases (Fig. 6C, D). Furthermore, AP

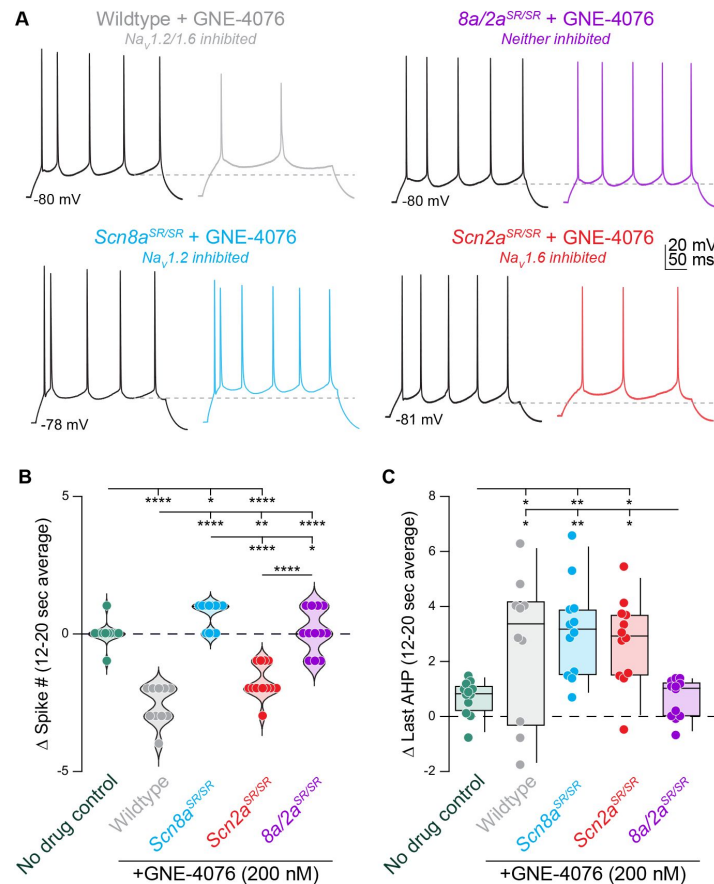


Figure 4

Acute inhibition of Na_v1.2 increases AP excitability

(A) AP train over 300 ms at baseline (black) or 18 sec post GNE-4076 onboarding (color) for all conditions (wildtype + GNE-4076, n=10; 8a/2a^{SR/SR} + GNE-4076, n=12; Scn2a^{SR/SR} + GNE-4076, n=12; Scn8a^{SR/SR} + GNE-4076, n=12). Dashed line represents V_m of last after-hyperpolarization (AHP).

(B) Summary data for Δ spike number at 12-20 sec post GNE-4076 onboarding. One-way ANOVA, Holm-Šídák multiple comparisons test. *p<0.05, **p<0.01, ****p<0.0001.

(C) Summary data for Δ last AHP at 12-20 sec post GNE-4076 onboarding. One-way ANOVA, Holm-Šídák multiple comparisons test. *p<0.05, **p<0.01.

(F) Recovery of AP peak velocity (dV/dt) represented as a delta value for individual cells plotted against time post GNE-4076 onboarding (log-scale). For Δ dV/dt, baseline value is subtracted from individual timepoints throughout the recovery phase (Δ V/s= recovery timepoint – baseline). Colors are matched to conditions represented in (B). Gray shaded bar represents recovery between 12-20 sec.

(G) Summary data for Δ dV/dt at 12-20 sec post GNE-4076 onboarding (time period represented as gray bar in (F)). Box plots show median and 90% tails. Circles represent individual cells. One-way ANOVA, Holm-Šídák multiple comparisons test.

****p<0.0001.

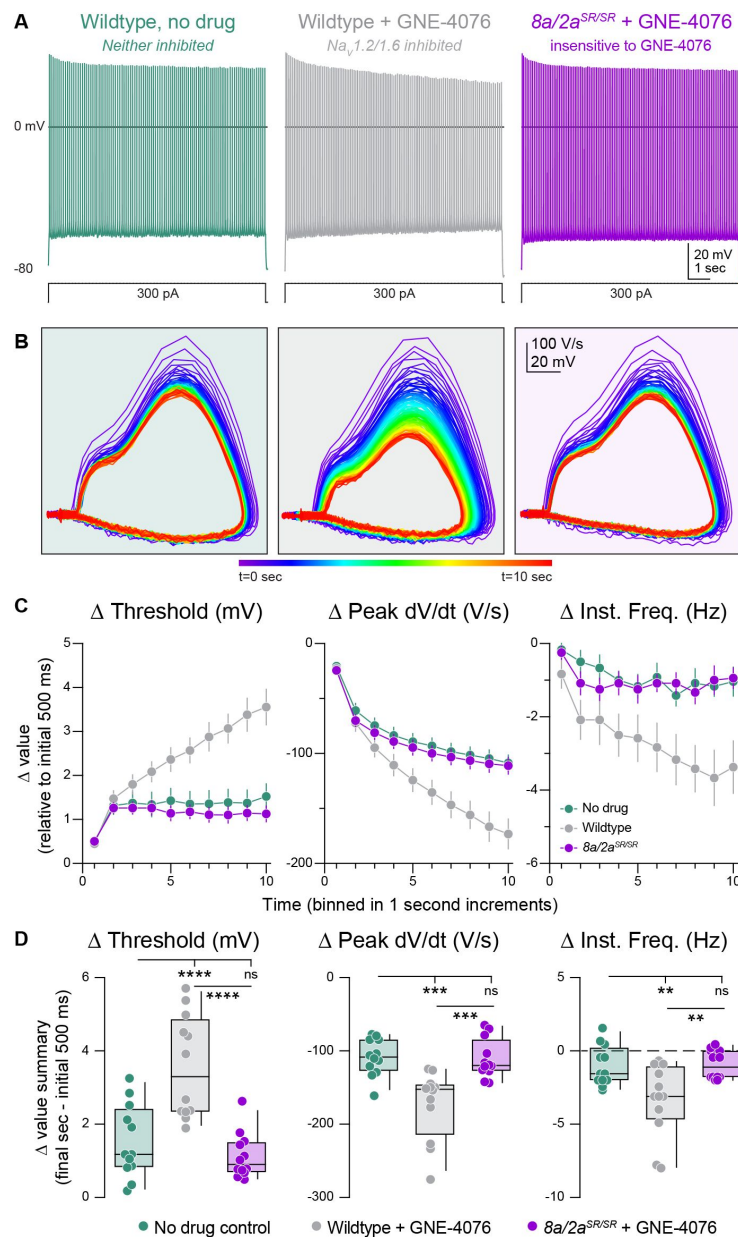


Figure 5

Activity-dependent onboarding of GNE-4076 to Na_v channels alters action potential firing properties in layer 5b, thick-tufted excitatory neurons

(A) Representative AP firing response to 300 pA current injection for 10 sec in wildtype or $8a/2a^{SR/SR}$ cells with or without 200 nM GNE-4076.

(B) Phase plane of AP traces shown in (A). Plots represent AP velocity by taking the first derivative (dV/dt, y-axis) versus the membrane potential (mV, x-axis). To represent changes with phase plane relative to time, a rainbow color spectrum is used with warmer colors representing more time lapsed.

(C) Delta threshold (Δ mV), delta peak dV/dt (Δ V/s) and delta instantaneous firing frequency (Δ Hz) binned in 1 sec increments normalized to the initial 500 ms of current injection (binned time – initial 500 ms). Circles represent mean Δ value $\pm$ SEM. Two-way ANOVA, Holm-Šidák multiple comparisons test.

(D) Summary data for the final sec in (C). Delta values are normalized to the initial 500 ms of the stimulus (binned time – initial 500 ms). Box plots show median and 90% tails. Circles represent individual cells (wildtype no drug, $n=12$; wildtype + GNE-4076, $n=12$; $8a/2a^{SR/SR}$ + GNE-4076, $n=12$). One-way ANOVA, Holm-Šidák multiple comparisons test. ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$.

frequency and peak somatic AP dV/dt was reduced to levels observed in WT cells (**Fig. 6C, D**). In cases where Na_v1.2 alone could be inhibited, changes in AP threshold and instantaneous frequency were no different than $8a/2a^{SR/SR}$ conditions. Indeed, the only change in AP properties was a decrease in peak AP dV/dt (**Fig. 6D**). This suggests that AP threshold and instantaneous frequency are more sensitive to Na_v1.6 antagonism, whereas peak somatic dV/dt can be affected by antagonism of either isoform. Absolute values are also reported for threshold, peak dV/dt and instantaneous frequency for all recorded cells in prolonged activity burst experiments (Fig. S5).

Leveraging use-dependent, isoform-selective Na_v pharmacology as anticonvulsants

Epilepsy is often associated with aberrant, excessive excitability in neocortical networks, whether from direct hyperexcitability in pyramidal cells^{53,54} or disinhibition of pyramidal cells via alterations in inhibitory networks^{37,55}. Several use-dependent Na antagonists with structures similar to GNE-4076 are being developed, exhibiting differential selectivity for Na_v 1.2 and Na_v 1.6²⁵. Use-dependence may have advantages as antiepileptics. In theory, they would have minimal effect on neurotypical network activity unless such networks were hyperactive enough to promote drug binding. This may occur preferentially in seizure states. But given results above demonstrating that block of Na_v1.2 and Na_v1.6 can have different effects on overall AP output, it is critical to determine how block of either channel affects overall activity in seizure-like conditions.

To test this, we mimicked synaptic input by repeatedly injecting a 60 sec-long somatic current composed of Poisson-distributed EPSC and IPSC-like waveforms designed to elicit ~10 Hz spiking in baseline conditions (**Fig. 7A**). Following this baseline, the PSC-like train was repeated, imposed upon a 400 pA standing current injection to increase spiking to ~30 Hz (**Fig. 7C**), mimicking spike rates commonly observed during cortical seizure^{56,57}. After this seizure-like event, cells were returned to baseline voltages and recovery was assessed with 4 more repetitions of the 60 sec PSC stimulus (**Fig. 7A-C**).

In the presence of 200 nM GNE-4076, baseline firing rates were mostly stable across all conditions (Fig. S6A). Wildtype, no drug and $8a/2a^{SR/SR}$ conditions observed no change in firing rate, while dual or selective Na_v1.6 blockade saw a slight decrease by 60 sec of postsynaptic activity (Fig. S6B). Interestingly, we did see a slight increase in baseline firing by 60 sec when Na_v1.2 was selectively targeted (Fig. S6A, B), furthering highlighting a paradoxical shift to hyperactive neurons when Na_v1.2 availability decreases. By contrast, a slight depolarization in AP threshold was observed at baseline when both channels were sensitive to GNE-4076 (e.g., WT plus drug) and to a lesser extent with selective block of Na_v1.6 (Fig. S6A, B). Given that this dose can completely block both Na_v1.2 and Na_v1.6 in HEK cells, a small amount of drug block was expected at baseline.

Seizure-like activity induction depolarized AP threshold markedly in all genotypes, and threshold depolarization persisted into the recovery phase for all neurons experiencing Na_v1.6 or dual inhibition (**Fig. 7C, D**). Seizure-like activity increased spike rate across all genotypes at the onset of seizure-like activity induction (**Fig. 7C, E**). All cells exhibited some degree of AP accommodation during this stimulus, including drug-naïve controls and cases where both Na_v1.2 and Na_v1.6 were insensitive to GNE-4076 block. WT cells exposed to GNE-4076 accommodated markedly, returning to baseline firing rates at the end of the seizure-like stimulus (**Fig. 7C, E**). A less dramatic effect was noted when Na_v1.6 was blocked alone (**Fig. 7E**). By contrast, cells where Na_v1.2 could be blocked alone were not appreciably different than untreated controls (**Fig. 7E**). Furthermore, some cells that remained hyperactive with selective Na_v1.2 block compared to controls cell throughout the recovery phase. Together, these data suggest that use-dependent pharmacology that targets Na_v1.2 and Na_v1.6 may be most beneficial if designed for higher potency at Na_v1.6.

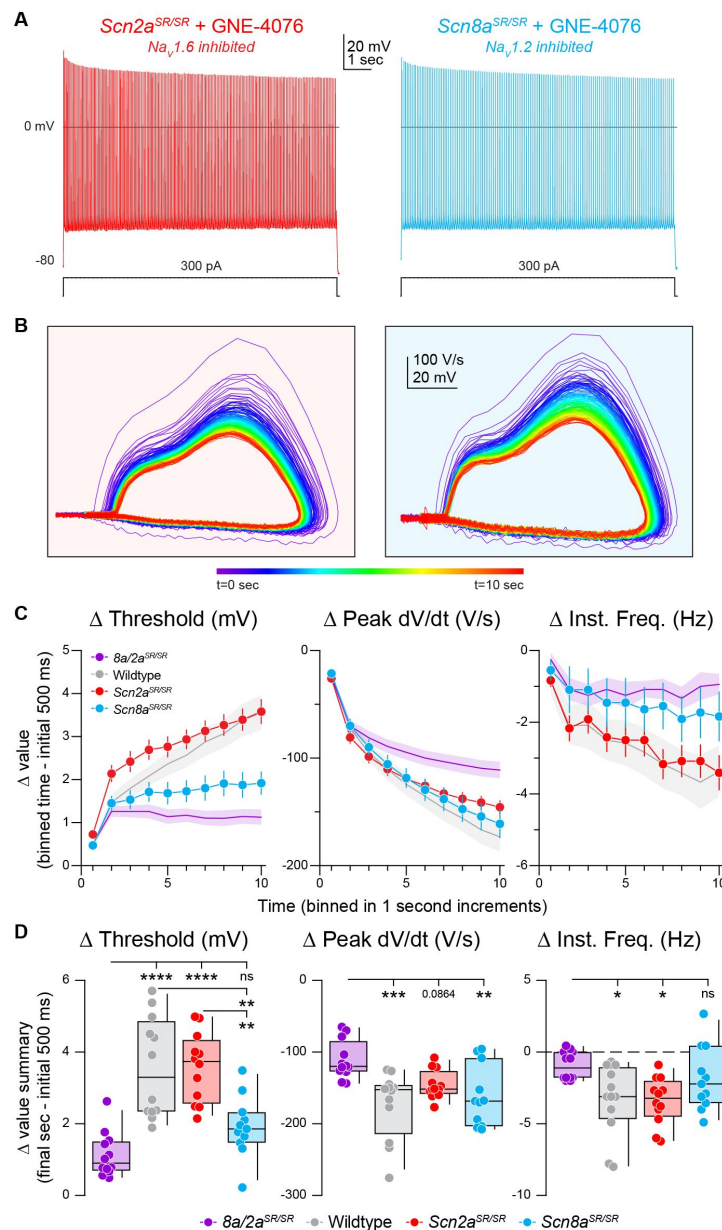


Figure 6

Selective inhibition of Na_v1.6 depolarizes AP threshold markedly while blocking both Na_v1.6 and Na_v1.2 reduces AP velocity

(A) Representative AP firing response to 300 pA current injection for 10 sec in *Scn2a^{SR/SR}* or *Scn8a^{SR/SR}* cells with 200 nM GNE-4076 to selectively inhibit Na_v1.6 or Na_v1.2, respectively.

(B) Phase plane of AP traces shown in (A). Plots represent AP velocity by taking the first derivative (dV/dt, y-axis) versus the membrane potential (mV, x-axis). To represent changes with phase plane relative to time, a rainbow color spectrum is used with warmer colors representing more time lapsed.

(C) Delta threshold (Δ mV), delta peak dV/dt (Δ V/s) and delta instantaneous firing frequency (Δ Hz) binned in 1 sec increments normalized to the initial 500 ms of current injection (binned time – initial 500 ms). Circles represent mean Δ value ± SEM. Average Δ value ± SEM for *8a/2a^{SR/SR}* + GNE-4076 and wildtype + GNE-4076 from Fig. 2D are represented.

(D) Summary data for the final sec in (C). Delta values are normalized to the initial 500 ms of the stimulus (binned time – initial 500 ms). Box plots show median and 90% tails. Circles represent individual cells (*8a/2a^{SR/SR}* + GNE-4076, n=12; wildtype + GNE-4076, n=12; *Scn2a^{SR/SR}* + GNE-4076, n=10; *Scn8a^{SR/SR}* + GNE-4076, n=12). One-way ANOVA, Holm-Šidák multiple comparisons test. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

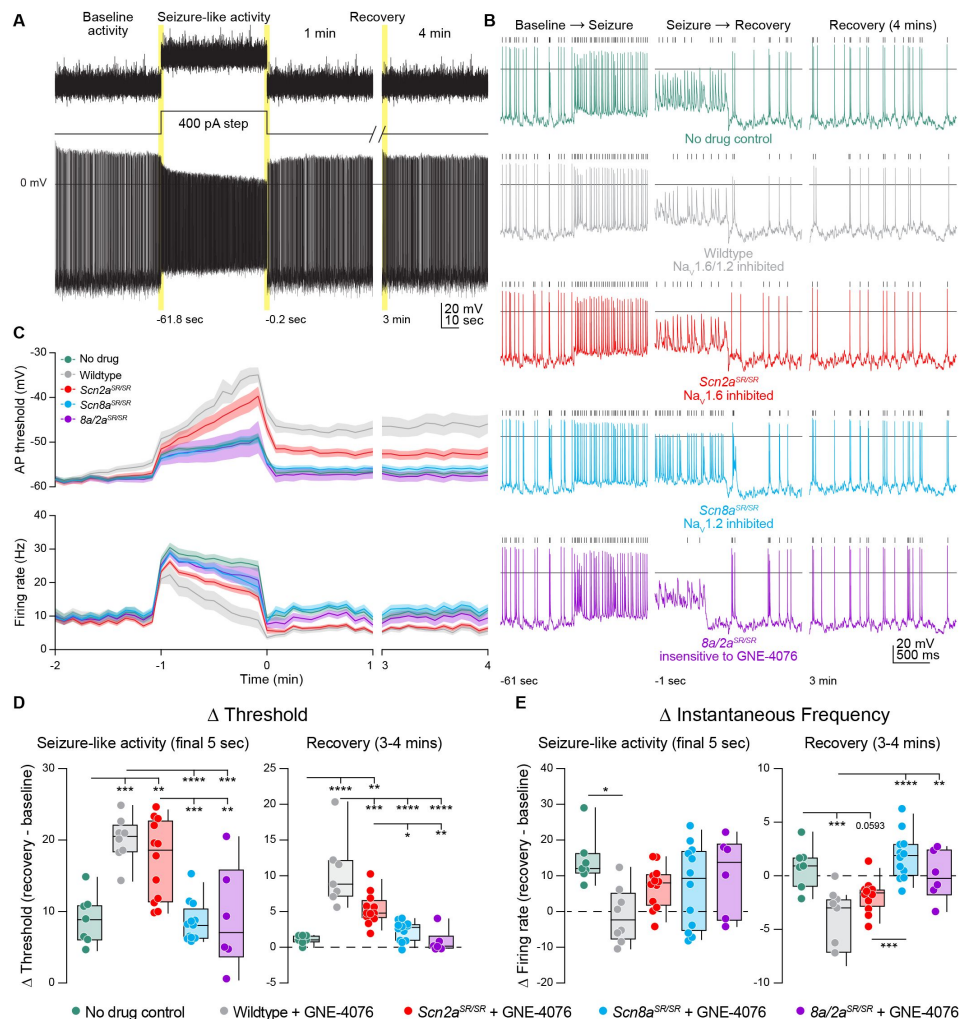


Figure 7

GNE-4076 onboarding following seizure-like activity continually impacts neuronal firing into recovery

(A) Stimulation protocol and example firing trace of cell injected with fluctuating post-synaptic potentials (PSPs) randomly generated using a Poisson probability distribution function for 60 sec. PSPs were continuously applied to acquire baseline activity, seizure-like activity and recovery activity. During seizure-like activity, a 400 pA step was applied in addition to the PSP. Recovery was continuously recorded for up to 4 mins post seizure-like activity.

(B) Zoomed-in example traces for all genotypes at the Baseline → Seizure transition, Seizure → Recovery transition and start of 3-4 min recovery period (highlighted in (A)). Solid horizontal black bar represents membrane potential (V_m) of 0 mV. Tick marks above traces represent detected spike defined as a change in V_m of 15 V/s or greater.

(C) Threshold (mV) or instantaneous firing frequency (Hz) binned in 5 sec increments for all genotypes in (B). Solid lines represent mean value $\pm$ SEM. Timescale on x-axis mirrors activity presented in (A).

(D) Summary of threshold data for the final 5 sec of seizure-like activity or entire 3-4 min recovery time-point in (C). Delta values are normalized to baseline activity (either at final 5 sec or entire period). Box plots show median and 90% tails. Circles represent individual cells (wildtype no drug, $n=7$; wildtype + GNE-4076, $n=7-8$; $Scn2a^{SR/SR}$ + GNE-4076, $n=12$; $Scn8a^{SR/SR}$ + GNE-4076, $n=12$; $8a/2a^{SR/SR}$ + GNE-4076, $n=6$). One-way ANOVA, Holm-Šidák multiple comparisons test. * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$.

(E) Summary of instantaneous frequency data for the final 5 sec of seizure-like activity or entire 3-4 min recovery time-point in (C). Delta values are normalized to baseline activity (either at final 5 sec or entire period). Box plots show median and 90% tails. Circles represent individual cells (wildtype no drug, $n=7$; wildtype + GNE-4076, $n=7-8$; $Scn2a^{SR/SR}$ + GNE-4076, $n=12$; $Scn8a^{SR/SR}$ + GNE-4076, $n=12$; $8a/2a^{SR/SR}$ + GNE-4076, $n=6$). One-way ANOVA, Holm-Šidák multiple comparisons test. * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$.

Discussion

Here, we combined genetic and pharmacological strategies to transiently, selectively and reversibly inhibit either Na_v1.6 or Na_v1.2 function using activity-dependent ASCs, which bind channels in the inactivated state. We show that acute blockade of either isoform has opposing effects on neuronal output: inhibition of Na_v1.6 decreases AP output, whereas Na_v1.2 increases AP output. Given this, we found that block of Na_v1.6 rather than Na_v1.2 was more effective at tempering spiking activity in cells driven to seizure-like levels of AP output. This suggests that pharmacology tuned to preferentially block Na_v1.6 over Na_v1.2 in a use-dependent manner may be useful as an anti-epileptic.

Functional implications of differential compartmental Na_v isoform expression

Electrophysiological recordings of AP propagation delays between somatic and axonal recordings demonstrate that APs initiate ~35 to 50 microns from the soma, in the distal AIS^{2,3,7,8}. This region is enriched with Na_v 1.6^{8,9,44,58,59}. Consistent with this, we find that inhibition of Na_v 1.6 alone alters AP initiation, with an increase in AP threshold and decrease in total AP number from block of this isoform. Following distal AIS AP initiation, APs sequentially propagate across different neuronal compartments to influence local excitability^{2,3,7}. Forward propagation was not examined here, but prior reports demonstrate that it is supported almost exclusively by Na_v 1.6 in nodes of Ranvier and boutons^{9,13,14}. By contrast, backpropagation through the proximal AIS and soma recruits a mix of Na_v1.6 and Na_v1.2.

Prior work has suggested that Na_v1.2 channels localized to the proximal AIS are critical for backpropagation, with modeling suggesting that APs would fail to effectively backpropagate into the soma in the absence of proximal AIS Na 1.2 channels⁸. While some efforts have been made to test this empirically with conditional *Scn2a* knockout, such experiments are imperfect, as Na_v1.6 redistributes in the AIS of cells that lack Na_v 1.2⁴. Here, we were able to test the role of Na_v 1.2 in backpropagation without compensatory effects, using acute pharmacological inhibition. Backpropagation failures were not observed, consistent with results from conditional *Scn2a* knockout⁴. However, we did observe a reduction in the velocity at which the AIS component of the AP depolarized when Na_v1.2 was blocked selectively (Fig. S3B-C). This suggests that proximal AIS-localized Na_v1.2 do aid in boosting backpropagating APs as they transit from the distal AIS to the soma. Nevertheless, Na_v1.2 channels are still dispensable for recruitment of somatic Na_v channels, at least in mouse mPFC layer 5b pyramidal cells. Whether similar effects are observed in other cells, including those with larger somata that may be more difficult to depolarize in the absence of Na_v1.2, remains to be tested.

Na_v1.2 expression in pyramidal cell dendrites appears to have two roles in neuronal excitability. Intuitively, these channels provide local inward current that boosts dendritic excitability, leading to a recruitment of voltage-gated calcium channels^{5,6}. But this depolarization also appears to be critical for recruitment of dendritically localized voltage-gated potassium channels that contribute to AP repolarization and net membrane potential between spikes⁴. Indeed, conditional knockout of Na_v 1.2 paradoxically increases AP output, and we suggested previously that this was due in large part to loss of interactions between dendritic Na_v channels and K_v channels. But given observed changes in Na_v1.6 function as well as potential for other compensatory changes in non Na_v ion channels, it was difficult to ascribe hyperexcitability purely to the loss of Na 1.2 alone⁵⁰. Here, we showed that identical increases in excitability—associated with depolarization of membrane potential between APs—could be observed with acute Na_v1.2

block. This indicates that such effects can be due purely to the interplay between dendritic $\text{Na}_V1.2$ and K_V channels, and that modifications to potassium channel distribution or function are not necessary for such effects.

Similar to $\text{Na}_V1.2$ conditional knockout, where $\text{Na}_V1.6$ is upregulated, knockout of $\text{Na}_V1.6$ results in an increase in $\text{Na}_V1.2$ expression, at least in the AIS. In pyramidal cells from mice constitutively lacking $\text{Na}_V1.6$, $\text{Na}_V1.2$ occupies the entirety of the AIS rather than just the region proximal to the soma^{10,16,44}. Thus, it has not been possible to evaluate the role of $\text{Na}_V1.6$ in its normal distribution using genetic manipulations. Here, we find that acute $\text{Na}_V1.6$ block reduces AP output with a concomitant increase in AP threshold. This contrasted markedly with block of $\text{Na}_V1.2$, which had no effect on threshold. Of note, this distinction can be leveraged to assess the specificity of other Na_V -targeting pharmacology that has been suggested to have specificity at select isoforms, as any change in threshold indicates that drugs are interacting with $\text{Na}_V1.6$ ^{10,21}.

ASC pharmacology for suppression of neuronal hyperexcitability

Epilepsy can arise from genetic and non-genetic factors, often with unexplained etiology^{60,61}. Seizure onset is broadly classified as an electrical imbalance of cellular and network activity that favors hyperexcitability⁶², whether it be cell intrinsic, synaptic, or due to complex network effects^{63,64}. Thus, seizure suppression can be targeted at multiple levels. Nevertheless, proper dosing can be difficult, as one aims to provide drug concentrations that temper excess activity but limit side effects like sedation associated with elevated drug concentrations.

For genetically defined sodium channelopathies, attention has been directed to neuronal cell-types that express each isoform (e.g., *SCN1A*: $\text{Na}_V1.1$, *SCN2A*: $\text{Na}_V1.2$ and *SCN8A*: $\text{Na}_V1.6$)^{65,66}. Seizures resulting from *SCN1A* loss of function limit inhibitory neuron excitability, thereby disinhibiting excitatory pyramidal cells^{37,55,67}. *SCN2A* gain-of-function results in hyperexcitability in pyramidal cells, especially in early development when $\text{Na}_V1.2$ channels are the sole isoform expressed in the AIS^{12,68}. Later in development, *SCN8A*-encoded $\text{Na}_V1.6$ channels are expressed more ubiquitously in the AIS of most cell classes, supporting AP initiation in both excitatory and inhibitory cells^{37,53,59,69}.

With these cellular distributions and mechanisms of action, guidelines have emerged for treatment. For *SCN1A* loss-of-function, sodium channel blocking anti-epileptics are typically counter-indicated^{70,71}. This is because currently prescribed sodium channel blockers are nonselective, with little preference for $\text{Na}_V1.1$, $\text{Na}_V1.2$ or $\text{Na}_V1.6$ ⁷². Thus, further block of the remaining $\text{Na}_V1.1$ leads to even more disinhibition of excitatory cells before any beneficial effects of blocking Na_V channels in excitatory cells are realized. Similarly, nonspecific sodium channel blockers are counter-indicated for *SCN2A* loss-of-function seizures, as they tend to increase seizure severity. Instead, non-selective Na_V inhibitors are useful when channel gain-of-function affects excitatory cells, as is the case for both *SCN2A* and *SCN8A* gain-of-function cases.

ASCs may be useful within each of these domains, as their chemistry can be adjusted to bias binding to specific isoforms in an activity-dependent manner^{24–27}. Indeed, compounds similar to those used here that inhibit both $\text{Na}_V1.2$ and $\text{Na}_V1.6$ (but not other Na_V channels) are effective at suppressing chemoconvulsant-induced seizures in *ex vivo* models²⁵. This parallels our observations, where excess activity was dampened most effectively by dual block of $\text{Na}_V1.2$ and $\text{Na}_V1.6$ (**Fig. 6C**). This effect appears due in large part to block of $\text{Na}_V1.6$, since, ultimately, this is the isoform responsible for AP initiation. Consistent with this, $\text{Na}_V1.6$ -preferring ASCs show promise in protecting from chemoconvulsant-induced seizures *in vivo*^{26,27}. Similar to our results, Johnson et al. also shows that a selective ASC for $\text{Na}_V1.6$ (NBI-921352) preferentially targets cortical pyramidal cells, unlike Carbamazepine that non-selectively targets many Na_V isoforms leading to impaired interneuron activity²⁷. In fact, drug therapies using agents like Carbamazepine, Lamotrigine and Phenytoin that have little to no selectivity for Na_V isoforms can have adverse effects in certain situations⁶². Thus, it may be that preferential block of $\text{Na}_V1.6$

would be beneficial in multiple sodium channelopathy conditions, including those associated with channel loss, since network hyperexcitability is ultimately dictated by $\text{Na}_v1.6$ function in the AIS of excitatory neurons after the first months of life.

Beyond selectivity, there may also be advantages to ASC activity-dependent properties. The ASC binding pocket is hidden from drug in a channel's closed state. This feature, combined with on- and off-rate kinetics, results in an accumulation of block primarily for highly-active neurons (Figs. 4, 6). Similar to on-demand, closed-loop electrical or optogenetic approaches for seizure intervention^{73–75}, ASCs are essentially biased to suppress prolonged, high-frequency activity, including the type of activity commonly observed during seizures. In this study, our focus centered mainly on seizure-like activity across individual neurons. Future studies can extend this approach to examining networks *ex vivo* and systems *in vivo*.

Methods

Resource availability

Lead contact

Any additional information or enquires related to reagents or resources should be directed to the lead contact, Kevin J. Bender (kevin.bender@ucsf.edu).

Materials availability

The transfer of unique reagents generated for this study will be made available upon request. A Materials Transfer Agreement may be required.

Data and code availability

This study did not generate any unique datasets or code. Data reported here will be made available by lead contact upon reasonable request.

Experimental models and subjects details

Mouse strains

All animal procedures are in accordance with the Institutional Animal Care and Use Committee (IACUC) guidelines in accordance with the University of California, San Francisco (UCSF). The following mouse strains were used in this study: C57BL/6J, YW- $\geq$ SR $\text{Na}_v1.2$ KI (*Scn2a*^{SR/SR}), YW->SR $\text{Na}_v1.6$ KI (*Scn8a*^{SR/SR}) and YW- $\geq$ SR dual KI (*8a/2a*^{SR/SR}). All experimental procedures were performed on mice maintained in-house on a 12:12 hour light-dark cycle under standard conditions with *ad libitum* access to food and water. For genotyping, genomic DNA was isolated from tail clip biopsies for PCR. Both male and female mice aged postnatal day (P)18-59 were used across all genotypes. C57BL/6J mice were obtained from Jackson Laboratories and YW- $\geq$ SR KI mice were developed by the Hackos Lab (Genentech).

Generation of SCN8A or SCN2A YW->SR KI mouse

As described previously²⁹, CRISPR/Cas9 technology^{30,31} was used to generate a genetically modified mouse strain with either an *Scn8a* or *Scn2a* YW->SR knock-in mutation. A single guide RNA (sgRNA) target and protospacer adjacent motifs (PAM) were identified for *Scn8a* ENSMUSG00000023033 or *Scn2a* ENSMUSG00000075318 genomic regions of interest using the CRISPR design tool (Benchling) that uses the algorithm described by Hsu et al.³² to provide 'MIT'

specificity scores for each sgRNA, as well as the top 15 predicted off-target loci and corresponding MIT off-target scores. The same guide target and PAM were used for both genes. Guide target: 5' CATTCTCTACTGGATTAATC 3'; PAM: TGG with an algorithm score of 42.3.

For the YW→SR mutation on the *Scn8a* gene, predicted cut sites are between 100,933,463-100,933,464 genome coordinates. The following oligonucleotide donor sequence was used: 5' ATGCTTATCTGCCTTAACATGGTGACCATGATGGTGGAGACAGACACA CAGAGCAAGCAGATGGAGAACATTCTCTCTCGGATTAATCTGGTCTTC GTCATCTTCTTCACCTGCGAGTGTGTGCTCAAAATGTTGCCTTGAGA CACTACTATTTC 3'. The first point mutation of Y1553S (TAC→TCT) is located at 100,933,454-100,933,456 genome coordinates, and a second point mutation of W1554R (TGG→CGG) is located at 100,933,457-100,933,459 genome coordinates.

For the YW→SR mutation on the *Scn2a* gene, predicted cut sites are between 166,900-166,901 genome coordinates. The following oligonucleotide donor sequence was used: 5' GAAATAGTAGTGTCTCAAGGCAAACATTTTGAG CACACACTCGCAGGTGAAGAAGATGACGAAGACCAGGTTGATCCGA GAGAGAATGTTCTCCATCTGCTTGCTCTGTGTGTCTGTCTCCACCATC ATGGTCACCATGTTAAGGCAGATAAGCAT 3'. The first point mutation of Y1553S (TAC→TCT) is located at 101035573-101035575 genome coordinates, and a second point mutation of W1554R (TGG→CGG) is located at 101035576-101035578 genome coordinates. Additionally, two silent mutations were created in the beginning of the gRNA to prevent Cas9 from cutting the donor oligo. The first silent mutation (ATT→ATC) is located at 101035579-101035581 genome coordinates and the second silent mutation (AAT→AAC) 101035582-101035584 genome coordinates.

After homology-directed repair of Cas9-induced chromosome breaks with the oligonucleotide donor, the YW→SR protein will be expressed. Once a sgRNA decision was finalized, the off-target list was used to identify the top 15, and next-generation sequencing (NGS) amplicon primers were designed for the on-target locus, and each of the off-targets synthetic guide RNA was obtained from Synthego. CAS9 protein was obtained from PROTEIN SOURCE and complexed with sgRNA before microinjection. Reagent concentrations for microinjection were as follows: 25 ng/μl Cas9 mRNA (Thermo Fisher; A29378) + 13 ng/μl sgRNA (Synthego), Oligonucleotide donor (50 ng/μl) (IDT).

After zygote microinjection and embryo transfer, genomic DNA was prepared from tail tip biopsies of potential G0 founders, and G0 animals were first analyzed by droplet digital PCR³³ (Bio-Rad). Primers were used to amplify the HDR event (ON Target) and the 15 most likely off-target sites. Only G0 mosaic founders positive for the intended mutation were screened by targeted amplicon NGS. Amplicons were submitted for NGS analysis.

Founders were selected for mating with wild-type C57BL/6N mice for germline transmission of the gene edited chromosome. Subsequent analysis of genomic DNA from G1 pups was used to confirm germline transmission of the targeted gene and the absence of off-target hits elsewhere in the genome.

To generate the dual *8a/2a*^{SR/SR} line, *Scn2a*^{SR/SR} and *Scn8a*^{SR/SR} mice were crossed and analysis of genomic DNA was used to confirm germline transmission of the targeted gene.

Method details

Synthesis of GNE4076

GNE-4076 was synthesized as previously described in Roecker et al.²⁸, and is is Compound 5 in original manuscript. We use GNE-4076 throughout this manuscript.

Generation of SCN2A YW->SR and SCN8A YW->SR constructs

The double mutation Y1564S/W1565R was introduced into the adult splice isoform of recombinant human Na_v1.2 (NCBI accession number NM_021007; AddGene #162279) using site directed mutagenesis as previously described³⁴. The corresponding mutations (Y1555S/W15556R) were engineered in recombinant human Na_v1.6 (adult isoform) as previously described³⁵. Mutagenic primer sequences are presented in Table 1. All plasmids were nanopore sequenced (Primoridium Labs, Arcadia, CA) to confirm the variants and exclude unwanted mutations.

Electrophysiology in immortalized cell lines

HEK293 cells were transfected with WT or mutant Na_v1.2 using the Invitrogen Lipofectamine LTX kit, whereas Na_v1.6 plasmids were electroporated into the LoNav derivative of ND7/23 cells using Maxcyte technology as previously described³⁵.

Na_v channel currents were recorded from HEK293 or ND7/23 cells using whole-cell patch clamp using a Molecular Devices Axopatch 200B amplifier. The recording pipet intracellular solution contained (in mM): 120 CsF, 10 NaCl, 2 MgCl₂, 10 HEPES, adjusted to pH 7.2 with CsOH. The extracellular recording solution contained (in mM): 155 NaCl, 3 KCl, 1 MgCl₂, 1.5 CaCl₂, 10 HEPES, adjusted to pH 7.4 with NaOH. Currents were recorded at 20 kHz sampling frequency and filtered at 5 kHz. Series resistance compensation was applied at 80%. Solutions containing GNE-4076 were applied using a Fluicell Dynaflo perfusion system.

We characterized dose response curves in both WT and mutant Na_v1.2 and Na_v1.6 channels by pulsing cells from -80 mV to 0 mV for 20 msec at a rate of 0.5 Hz to establish a baseline current. Cells were then perfused with GNE-4076 starting at 30 nM and increased up to 100 μM. To allow adequate GNE-4076 onboarding to Na_v isoforms/mutants at each successive dose, cells were depolarized to 0 mV for 10 sec followed by similar test pulses used to acquire baseline current. Between dose increases, cells were held at -80 mV to allow adequate unbinding of GNE-4076.

We also characterized the activation and steady-state inactivation properties of both WT and mutant Na_v1.2 and Na_v1.6 channels. To measure activation, we used a holding voltage of -80 mV, a short 20 ms pre-pulse to -120 mV, and a 30 ms pulse to voltages ranging from -100 mV to -20 mV in steps of 5mV at a rate of once per 3 sec. P/4 leak subtraction was used to reduce leak currents. The peaks of the resulting Na_v currents were measured. Conductance was calculated using the equation $G = I / (V - V_{Na})$, normalized, and plotted as a function of voltage. To measure inactivation, we started at a holding voltage of -120 mV to bring the channels fully into the closed state (about 1 min). We then pulsed to 0 mV while reducing the holding voltage from -120 mV to +30 mV in steps of 5 mV at a rate of once per 5 sec. The peaks of the resulting Na_v currents were measured, normalized and plotted as a function of voltage. To quantify these biophysical properties, we fit the activation and inactivation curves to the following Boltzmann equations:

$$\begin{aligned} G/G_{max} &= 1 - \frac{1}{1 + e^{(V - V_{1/2})/sf}} && \text{(activation)} \\ I/I_{max} &= \frac{1}{1 + e^{(V - V_{1/2})/sf}} && \text{(inactivation)} \end{aligned}$$

$V_{1/2}$ is the voltage where half-maximal peak conductance (or current) was observed, and sf is the slope factor.

Ex vivo electrophysiology

Mice aged P18-P59 were anesthetized with isoflurane prior to harvesting the brain. Dissected brains were immediately placed in cutting solution (4°C) containing 87 mM NaCl, 25 mM NaHCO₃, 25 mM glucose, 75 mM sucrose, 2.5 mM KCl, 1.25 mM NaH₂PO₄, 0.5 mM CaCl₂ and 7 mM MgCl₂ that

is bubbled with 5% CO₂ / 95% O₂. Coronal slices were prepared from the medial prefrontal cortex (PFC) at a thickness of 250 microns and placed in a holding chamber warmed to 33°C for 30 minutes. Slices were then allowed to recover at room temperature until recording. Recordings were performed at 31–33°C in a solution containing 125 mM NaCl, 2.5 mM KCl, 2 mM CaCl₂, 1 mM MgCl₂, 25 mM NaHCO₃, 1.25 mM NaH₂PO₄ and 25 mM glucose that is bubbled with 5% CO₂ / 95% O₂. Osmolarity of the recording solution was adjusted to approximately 309 mOsm.

Neurons were identified using differential interference contrast (DIC) optics for conventional visually guided whole-cell recordings. For current clamp experiments, pipettes were pulled from Schott 8250 glass, with a tip resistance of 3–4 M Ω , and filled with a K-gluconate based internal solution containing 113 mM K-Gluconate, 9 mM HEPES, 4.5 mM MgCl₂, 0.1 mM EGTA, 14 mM Tris₂-phosphocreatine, 4 mM Na₂-ATP, 0.3 mM Tris-GTP; 290 mOsm; pH: 7.2–7.25. All data were corrected for measured junction potentials of 12 mV.

All electrophysiology data were acquired through custom protocols generated in IgorPro (Wavemetrics) via Multiclamp 700A or 700B amplifiers (Molecular Devices). Action potential (AP) waveform measurements were acquired at 50 kHz and low-pass Bessel filtered at 20 kHz for all experiments except for data presented in **Fig. 6** (acquired at 10 kHz and filtered at 3 kHz). Pipette capacitance was compensated in all current-clamp recordings to 50% of the fast capacitance measured after membrane seals were established in voltage-clamp. The bridge was balanced, and series resistance was kept < 18 M Ω in all recording. Any cells with input resistance changes exceeding $\pm$ 15% were omitted from dataset. A quartz electrode holder (Sutter Instrument) was used to collect all recording to minimize physical drift of recording electrodes.

All recordings were made from medial prefrontal cortex (PFC) layer 5b, thick-tufted pyramidal tract (PT) neurons. Pyramidal neuron identity was confirmed by assessing membrane responses to hyperpolarizing current (–400 pA, 120 ms), with PT neurons defined as those that exhibit membrane depolarization overshoot that peaks within 90 ms of current step offset³⁶. AP threshold, AIS dV/dt and peak dV/dt measurements were determined from either the first AP or all APs within a spike train evoked by a stimulus. AP threshold was defined as the membrane potential (V_m) when the AP speed (dV/dt) exceeds 15 V/s. Somatic peak (V/s) was defined as the max value in the first derivative (dV/dt) of an AP. AIS max (V/s) was defined as the trough or saddle point in the second derivative (d²V/dt²) that occurs during the depolarizing phase of an AP before the somatic peak³⁷. AIS inflection point was defined as the peak of the second derivative (d²V/dt²) that occurs prior to the AIS max or somatic peak. Frequency was defined as the spike number in a train elicited per second (spikes/sec or Hz). Afterhyperpolarization (AHP) was defined as the minimum voltage between 2 spikes in a train.

Spike trains were evoked by injecting current (250–350 pA, 10 sec or 300 ms duration) in the presence or absence of 200 nM GNE-4076. In recovery experiments, neurons were held at –12 mV for 30 sec (voltage-clamp) to depolarize cells, activate Na_v channels and expose the ASC binding pocket to maximize GNE-4076 onboarding and Na_v inhibition. The interstimulus interval given during the recovery period was as follows: 2 sec, ~20–30 sec duration; 5 sec, ~30–45 sec duration; 15 sec, ~60–90 sec duration; 30 sec, ~120–180 sec duration; and 60 sec, ~60–240 sec duration. In experiments mimicking cellular activity, a post-synaptic potential train was randomly generated using a Poisson probability distribution function for 60 sec with a frequency of 50 Hz and amplitude standard deviation of 200 pA.

For nucleated patch voltage-clamp experiments, recordings were made at 31–33°C in a solution containing 125 mM NaCl, 2.5 mM KCl, 2 mM CaCl₂, 1 mM MgCl₂, 25 mM NaHCO₃, 1.25 mM NaH₂PO₄, 15 mM glucose, 4 mM TEA, 1 mM 4-AP, and 10 μ M nifedipine, with or without 200 nM GNE-4076, (bubbled with 5% CO₂ / 95% O₂). A Cs-methanesulfonate based recording solution was used: 110 mM CsMeSO₃, 40 mM HEPES, 1 mM KCl, 4 mM NaCl, 4 mM Mg-ATP, 10 mM Na-phosphocreatine, 0.4 mM Na₂-GTP, and 0.1 mM EGTA; 290 mOsm; pH: 7.2–7.25. Whole-cell

recordings were established before withdrawing pipettes from the slice, pulling a region of the somatic membrane with the pipette. Neurons were held to -80 mV and stepped to -12 mV every 2 sec, 5-10 times, then neurons were held to -12 mV for 10 sec to onboard GNE-4076, then returned to -80 mV and probed for Na_V recovery with steps to -12 mV every 2 sec. Leak currents were subtracted with a P/8 protocol using steps from -80 to -90 mV.

Compartmental modeling

A compartmental model was constructed within the NEURON environment to simulate a layer 5 pyramidal neuron as described before^{4,38,39}. A multi-compartmental model, originally developed by the Blue Brain Project, was implemented to reflect the morphology detailed by Ramaswamy and Markram⁴⁰ and electrophysiological features were adjusted to reflect empirically obtained data^{4,38}. We modified the model by replacing the aggregated sodium conductances (NaT and NaP) with distinct $\text{Na}_V1.2$ and $\text{Na}_V1.6$ channels to match empirically observed distributions⁸. $\text{Na}_V1.2$ and $\text{Na}_V1.6$ channels were distributed throughout the cell with equal levels in the soma and $20\text{ }\mu\text{m}$ of the proximal dendrites. $\text{Na}_V1.2$ was solely expressed in dendrites more distal to the soma, inferred based on AP-evoked sodium imaging observations in *Scn2a*^{+/−} conditions⁶.

The two axonal compartments were subdivided into an axon initial segment and distal axon. Within the AIS, $\text{Na}_V1.2$ and $\text{Na}_V1.6$ were distributed with increased $\text{Na}_V1.2$ in the proximal AIS and increased $\text{Na}_V1.6$ in the distal AIS to recapitulate the channel distribution as previously observed empirically⁸. $\text{Na}_V1.2$ was not included in the distal AIS or axon where only $\text{Na}_V1.6$ is present, including an enriched region to model a node of Ranvier. To simulate blocking of $\text{Na}_V1.2$ and $\text{Na}_V1.6$, each channel's density was globally reduced in 10% increments from 100% to 0%. Both $\text{Na}_V1.2$ and $\text{Na}_V1.6$ channels were represented using the Hodgkin-Huxley formalism⁴¹. Parameter optimization for both channels was conducted using an evolutionary algorithm from BluePyOpt⁴² and adapted for use with the computational resources at the National Energy Research Computing Center similar to Ladd et al.⁴³.

To validate parameter sensitivity, we constructed multiple variations of our model with different distributions of $\text{Na}_V1.2$ and $\text{Na}_V1.6$ in the AIS. The crossover point at which the $\text{Na}_V1.2$ and $\text{Na}_V1.6$ distribution curves intersect was shifted distally in increments of $3.75\text{ }\mu\text{m}$ to make 5 variations of the AIS with increasing density of $\text{Na}_V1.2$. With the wildtype crossover point located $15\text{ }\mu\text{m}$ distal to the soma, the 5 right shifted variations have crossover points at $18.75\text{ }\mu\text{m}$, $22.5\text{ }\mu\text{m}$, $26.25\text{ }\mu\text{m}$, $30\text{ }\mu\text{m}$, and $33.75\text{ }\mu\text{m}$ respectively. Two variations of the AIS with increased $\text{Na}_V1.6$ density were created by shifting the crossover point proximally in $3.75\text{ }\mu\text{m}$ increments to $11.25\text{ }\mu\text{m}$ and $7.5\text{ }\mu\text{m}$. To see how our model would behave with different AIS sodium channel distributions, we varied the ratio of $\text{Na}_V1.2$: $\text{Na}_V1.6$ density in the whole cell to test different conditions at each AIS crossover point and measured the resulting threshold, AIS peak, and somatic peak of the phase plane plot. $\text{Na}_V1.2$ percentage was decreased from 90% to 0% in 10% increments as $\text{Na}_V1.6$ percentage was concurrently increased from 10% to 100% in 10% increments. The $\text{Na}_V1.2$: $\text{Na}_V1.6$ ratio of 100%:0% did not produce any action potentials and was therefore not included in the analysis. In instances where the somatic portion of the phase plane was not a true peak that consisted of adjacent values less than a local maxima, the prominence was estimated by taking the index of the second derivative value closest to zero and calculating the dV/dt value at that index.

Quantification and statistical analysis

Data are reported as absolute values or the absolute difference from baseline (Δ). For Δ values, data were normalized either to the initial 500 ms of the stimulus (**Fig. 2–3**) or baseline spiking (**Fig. 4–5**). Time course graphs are represented as a mean $\pm$ standard error. Summary graphs are represented with boxplots showing the median, quartiles and 90% tails or with violin plots. All summary graphs overlay individual datapoints and represent recordings from single

cells (reported n) for all electrophysiology experiments. Data were obtained from 5-9 animals (both sexes) per condition, which are standard group sample sizes used in the field. Analysis was performed blind to genotype $\pm$ drug. Statistical analysis was performed using Prism 10 (Graphpad Software). Quantified mean $\pm$ standard error and statistical test used is noted in figure legends. Significance was set at an alpha value of 0.05 and 'ns' indicates no significance.

Author Contributions

DHH and KJB conceived the project. JDG, CW, RA, EB, TF, JMD, TVA, ALG, RBS, DDH and KJB designed and performed experiments. JDG, CW, RA, EB, TF, JMD, TVA, ALG, RBS, DDH and KJB analyzed the data. JDG and KJB wrote the manuscript. EB, TF, ALG, RBS, and DDH reviewed and edited the manuscript.

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Additional files

Supplemental tables and figures [🔗](#)

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Reviewer #1 (Public review):

Summary:

Prior research indicates that Nav1.2 and Nav1.6 have different compartmental distributions, expression timelines in development, and roles in neuron function. The lack of subtype-specific tools to control Nav1.2 and Nav1.6 activity however has hampered efforts to define the role of each channel in neuronal behavior. The authors attempt to address the problem of subtype specificity here by using aryl sulfonamides (ASCs) to stabilize channels in the inactivated state in combination with mice carrying a mutation that renders Nav1.2 and/or Nav1.6 genetically resistant to the drug. Using this innovative approach, the authors find that action potential initiation is controlled by Nav1.6 while both Nav1.2 and Nav1.6 are involved in back-propagation of the action potential to the soma, corroborating previous findings. Additionally, Nav1.2 inhibition paradoxically increases firing rate, as has also been observed in genetic knockout models. Finally, the potential anticonvulsant properties of ASCs were tested. Nav1.6 inhibition but not Nav1.2 inhibition was found to decrease action potential firing in prefrontal cortex layer 5b pyramidal neurons in response to current injections designed to mimic inputs during seizure. This result is consistent with studies of loss-of-function Nav1.6 models and knockdown studies showing that these animals are resistant to certain seizure types. These results lend further support for the therapeutic promise of activity-dependent, Nav1.6-selective, inhibitors for epilepsy.

Strengths:

- (1) The chemogenetic approaches used to achieve selective inhibition of Nav1.2 and Nav1.6 are innovative and help to resolve long-standing questions regarding the role of Nav1.2 and Nav1.6 in neuronal electrogenesis.
- (2) The experimental design is overall rigorous, with appropriate controls included.
- (3) The assays to elucidate the effects of channel inactivation on typical and seizure-like activity were well selected.

Weaknesses:

(1) As discussed in the revised manuscript, the fact that channels are only partially blocked by the ASC and that ASCs act in a use-dependent manner complicates the interpretation of the effects of Nav1.2 versus Nav1.6 on neuronal activity.

(2) The idea that use-dependent VGSC-acting drugs may be effective antiseizure medications is well established. Additional discussion of the existing, widely used, use-dependent VGSC drugs (e.g. Carbamazepine, Lamotrigine, Phenytoin) would improve the manuscript. Also, the idea that targeting Nav1.6 may be effective for seizures is established by studies using genetic models, knockdown, and partially selective pharmacology (e.g. NBI-921352). Additional discussion of how the results reported here are consistent with or differ from studies using these alternative approaches would improve the discussion.

<https://doi.org/10.7554/eLife.105696.2.sa2>

Reviewer #2 (Public review):

The authors used a clever and powerful approach to explore how Nav1.2 and Nav1.6 channels, which are both present in neocortical pyramidal neurons, differentially control firing properties of the neurons. Overall, the approach worked very well, and the results show very interesting differences when one or the other channel is partially inhibited. The experimental data is solid and the experimental data is very nicely complemented by a computational model incorporating the different localization of the two types of sodium channels.

The revised manuscript has re-organized figures that make the results and interpretation easier to follow.

<https://doi.org/10.7554/eLife.105696.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

Prior research indicates that Nav1.2 and Nav1.6 have different compartmental distributions, expression timelines in development, and roles in neuron function. The lack of subtype-specific tools to control Nav1.2 and Nav1.6 activity however has hampered efforts to define the role of each channel in neuronal behavior. The authors attempt to address the problem of subtype specificity here by using aryl sulfonamides (ASCs) to stabilize channels in the inactivated state in combination with mice carrying a mutation that renders Nav1.2 and/or Nav1.6 genetically resistant to the drug. Using this innovative approach, the authors find that action potential initiation is controlled by Nav1.6 while both Nav1.2 and Nav1.6 are involved in backpropagation of the action potential to the soma, corroborating previous findings. Additionally, Nav1.2 inhibition paradoxically increases the firing rate, as has also been observed in genetic knockout models. Finally, the potential anticonvulsant properties of ASCs were tested. Nav1.6 inhibition but not Nav1.2 inhibition was found to decrease action potential firing in

prefrontal cortex layer 5b pyramidal neurons in response to current injections designed to mimic inputs during seizure. This result is consistent with studies of loss-of-function Nav1.6 models and knockdown studies showing that these animals are resistant to certain seizure types. These results lend further support for the therapeutic promise of activity-dependent, Nav1.6-selective, inhibitors for epilepsy.

Strengths:

- (1) The chemogenetic approaches used to achieve selective inhibition of Nav1.2 and Nav1.6 are innovative and help resolve long-standing questions regarding the role of Nav1.2 and Nav1.6 in neuronal electrogenesis.*
- (2) The experimental design is overall rigorous, with appropriate controls included.*
- (3) The assays to elucidate the effects of channel inactivation on typical and seizure-like activity were well selected.*

Weaknesses:

- (1) The potential impact of the YW->SR mutation in the voltage sensor does not appear to have been sufficiently assessed. The activation/inactivation curves in Figure 1E show differences in both activation and inactivation at physiologically relevant membrane voltages, which may be significant even though the V1/2 and slope factors are roughly similar.*

We have performed new experiments testing how YW->SR mutations affect spiking on their own. The reviewer's intuition was correct; the small changes in voltage-dependence in Nav1.6 identified in heterologous expression systems translated into a ~2 mV hyperpolarization in threshold in neurons.

- (2) Additional discussion of the fact that channels are only partially blocked by the ASC and that ASCs act in a use-dependent manner would improve the manuscript and help readers interpret these results.*

We have updated text extensively to address this concern. Details are found in the author suggestions below.

- (3) Nav1.6 was described as being exclusively responsible for the change in action potential threshold, but when Nav1.6 alone was inactivated, the effect was significantly reduced from the condition in which both channels were inactivated (Figure 4E). Similarly, Figure 6C shows that blockade of both channels causes threshold depolarization prior to the seizure-like event, but selective inactivation of Nav1.6 does not. As Nav1.2 does not appear to be involved in action potential initiation and threshold change, what is the mechanism of this dissimilarity between the Nav1.6 inactivation and combined Nav1.6/ Nav1.2 inactivation?*

We believe the dissimilarity is due to interactions between Nav1.2 and other channel classes (e.g., potassium channels) throughout the cell, including the somatodendritic domain. Nav1.6 that initiates APs, localized to the AIS, do not live in isolation, and AP threshold can be affected by the recent membrane potential history. Loss of Nav1.2-mediated depolarization in the dendrites begets less potassium channel-mediated repolarization, as described in Figure 4.

- (4) The idea that use-dependent VGSC-acting drugs may be effective antiseizure medications is well established. Additional discussion or at least acknowledgement of the existing, widely used, use-dependent VGSC drugs should be included (e.g.*

Carbamazepine, Lamotrigine, Phenytoin). Also, the idea that targeting Nav1.6 may be effective for seizures is established by studies using genetic models, knockdown, and partially selective pharmacology (e.g. NBI-921352). Additional discussion of how the results reported here are consistent with or differ from studies using these alternative approaches would improve the discussion

We agree; the concept of use-dependent block as a means to treat seizure is not new, and we have updated the discussion to include commentary on other medications currently in use. What is new here is our ability to explore the role of Nav1.2 and Nav1.6 in electrogenesis with a level of drug selectivity that could not be achieved without the addition of the YW->SR mutations. This approach in itself will not be useful in the clinic, but it may help guide drug design in the future. One major interpretation of this work is that Nav1.6 block is more effective than Nav1.2 block in general, and may even be effective for non-SCN8A genetic conditions. This is indeed one of the reasons that we believe that drugs like NBI-921352, itself an aryl-sulfonamide, is being tested in seizure models.

Reviewer #2 (Public review):

The authors used a clever and powerful approach to explore how Nav1.2 and Nav1.6 channels, which are both present in neocortical pyramidal neurons, differentially control firing properties of the neurons. Overall, the approach worked very well, and the results show very interesting differences when one or the other channel is partially inhibited. The experimental data is solid and the experimental data is very nicely complemented by a computational model incorporating the different localization of the two types of sodium channels.

In my opinion the presentation and interpretation of the results could be improved by a more thorough discussion of the fact that only incomplete inhibition of the channels can be achieved by the inhibitor under physiological recording conditions and I thought the paper could be easier to digest if the figures were re-organized. However, the key results are well-documented.

This is a concern raised by multiple reviewers, and we thank you all for your help in improving the way in which we discuss the results. We have revised the manuscript extensively, moving figures around per your advice and the advice of R1 in their comments to authors.

Reviewer #3 (Public review):

Summary:

The authors used powerful and novel reagents to carefully assess the roles of the voltage gated sodium channel (Nav) isoforms in regulating the neural excitability of principal neurons of the cerebral cortex. Using this approach, they were able to confirm that two different isoforms, Nav1.2 and Nav1.6 have distinct roles in electrogenesis of neocortical pyramidal neurons.

Strengths:

Development of very powerful transgenic mice in which Nav1.2 and/or Nav1.6 were modified to be insensitive to ASCs, a particular class of Nav blocker. This allowed them to test for roles of the two isoforms in an acute setting, without concerns of genetic or functional compensation that might result from a Nav channel knockout.

Careful biophysical analysis of ASC effects on different Nav isoforms.

Extensive and rigorous analysis of electrogenesis - action potential production - under conditions of blockade of either NaV1.2 or NaV1 or both.

Weaknesses:

Some results are overstated in that the representative example records provided do not directly support the conclusions.

We have swapped out example records to better capture the median effect observed and to better capture our discussion of these results. Please see below, in recommendations for authors, for details.

Results from a computational model are provided to make predictions of outcomes, but the computational approach is highly underdeveloped.

Modeling has been elaborated upon extensively, with more detail in methods, a new sensitivity analysis supplemental figure, and a deposition into ModelDB. Please see below, in recommendations for authors, for details.

Reviewer #1 (Recommendations for the authors):

Regarding the concern about the potential impact of the YWàSR mutation: All results in Figures 2-6 report only within-subject changes before and after drug-activating protocols. These results show that the drug has no effect on the mutant channel, but whether the mutant channel itself has any effect on neuronal properties is not clear. This deficiency could be rectified by reporting raw values for AP threshold, spike rate, etc. in the pre-drug condition and statistically analyzing the apparent differences in the activation/inactivation curves.

Data in our original submission only included data in the presence of GNE-4076. We now present new data showing how the YWàSR mutation affects baseline activity of neurons. These data are in Supplemental Figure 1. Compared to wildtype (no drug control) neurons, we observe no change in peak dV/dt. However, threshold is hyperpolarized by approximately 2 mV in dual knockin neurons (median values: -57.4 mV for dual knockin and -55 mV for wildtype). This is consistent with measures from heterologously expressed channels, where we observed somewhat subtle shifts in voltage-dependence of inactivation and activation in NaV1.6 as a result of YWàSR incorporation.

In addition to these data, we also include the baseline dataset from Figure 3, where GNE-4076 is present throughout recording, and report that neither threshold nor peak dV/dt are influenced by the presence of GNE at baseline. This suggests that any drug binding at baseline (i.e., before firing APs via somatic current injection) is negligible, consistent with the concept that GNE-4076 has low affinity for the closed channel state.

Minor Comments:

While the single-cell response to "seizure-like" input aptly demonstrates the change in action potential threshold and firing rate induced by NaV1.6 inhibition, this component of the paper could be enhanced by a network-level assay that assesses the impact of this drug on an actual seizure-like event in acute slices or on seizure susceptibility in vivo.

This is an excellent thought, and the work near the end of this manuscript is an effort to mimic network-like activity in a controlled way in single cells. To expand this to bona fide seizure-like activity in acute slices or *in vivo* is something that we are considering for future

studies. To do this properly requires extensive validation of dosing and seizure induction that will require several years' effort.

| *Fig 1e caption says "circles" but the markers are squares*

This has been corrected, thank you for catching it.

| *Color scheme in S2B is not intuitive to me*

We've now updated the caption to better describe the color scheme used within.

| *Fig S2: graph or show change in threshold*

Empirical threshold data are in main figure 3D. Changes in threshold related to modeling are now included in a new sensitivity analysis that is in a new Supplemental Figure 2.

| *Fig 3A example of Nav1.6 inhibition does not show change in AP threshold apparent in the aggregate data*

We have updated the representative example to better illustrate the change in AP threshold for Nav1.6 inhibition.

| *"AP initiation is mediated exclusively by Nav1.6" not corroborated by data; APs still occur when Nav1.6 is inhibited*

This was an over-interpretation of our data, indeed. We have updated the language to be more accurate to the following: "AP threshold and AP initiation appears to be initiated in an Nav1.6-rich region in control conditions; when Nav1.6 is inhibited, APs can occur at more depolarized potentials, likely mediated predominately by Nav1.2."

| *Fig S3C missing WT/Scn8aSR/SR significance marking. Chosen example makes it look like there is a small decrease.*

Please note that there is no difference between these two conditions when in delta dV/dt for AIS inflection point ($p = 0.4344$).

Reviewer #2 (Recommendations for the authors):

This manuscript presents a clever and powerful approach to examining differential roles of Nav1.2 and Nav1.6 channels in excitability of pyramidal cell excitability, by engineering mice in which a sulfonamide inhibitor of both channels has reduced affinity for one or the other. Overall, the results in the manuscript are interesting and give important information about differential roles of Nav1.6 and Nav1.2 channels.

The paper makes an important contribution to better understanding distinct roles of Nav1.2 and Nav1.6 channels. This improved understanding could help guide design of anti-seizure drugs targeted to sodium channels.

Having made it clear that I think this is an important and impressive piece of work for which the authors should be congratulated, I found reading and interpreting the manuscript a frustrating experience. I will be blunt about the ways in which I found the presentation and discussion to be frustrating and even annoying, in the spirit of frank feedback by one interested and appreciative reader that the authors can consider or reject as they wish.

From the start, I had the feeling that the authors were presenting and discussing the results in a sanitized "never-mind-about the details" fashion such as might be appropriate for a seminar to a general audience not interested in details, but not appropriate for a research paper.

Our intent certainly was not to frustrate or annoy readers. We are very grateful that you have provided these comments, which have certainly improved the manuscript, hopefully mitigating some of the frustration for future readers. We appreciate that there are complex drug and voltage effects occurring within these studies, and in an effort to distill these effects into digestible prose, we appear to have been too earnest. We have expanded on the requested topics below and please note that, for the aficionados, every figure displays individual data. Further, we have made a special effort to ensure that features of excitability are presented throughout the drug and manipulation timecourse, including time-points before and after periods subject to statistical comparison, so that the reader may draw their own conclusions.

General:

There were two major ways in which I found the presentation and discussion frustrating and even annoying: First, not clearly discussing early in the presentation the fact that it is impossible to achieve complete inhibition with this agent during measurements of physiological firing and second, presenting so much of the effects as deltas of various parameters rather than showing effects on absolute values of the parameters.

Our response to the first issue will follow the next comment, as it relates to this statement. Regarding use of deltas and absolute values for changes in threshold and dV/dt across figures. Every cell has a unique AP threshold and peak dV/dt , and we found that displaying data zeroed to baseline values best illustrated the effects of GNE-4076. Without this, GNE-based effect could be buried within the cell-to-cell variability. This helped most when trying to make the case that threshold was unaffected in 2a/8a YWàSR knockin animals. We continue to believe that this is the best way to display the data in the primary figures, but to provide a more complete account, we now present absolute values in supplemental tables and supplemental figures.

The first issue, the incomplete inhibition by the agent, was the most annoying because the authors obviously thought a lot about this and even closed the paper by proposing this as a positive feature of this class of inhibitors, yet discussed it only piecemeal - and with most of the key experimental data in the Supplement. There are two fundamental characteristics of this (and other) sulfonamide inhibitors that complicate interpretation of experiments, especially when applied in a slice experiment: they only bind to the channel when the channel is depolarized, and even when the channel is depolarized for many seconds, bind very slowly to the channel.

That makes it almost impossible to know exactly what fraction of channels is being inhibited during measurements of firing. Obviously, the authors are well-aware of this issue and they allude to it and even make use of it in some of the protocols, but they never really discuss it in a very clear manner.

We agree that it is impossible to know the precise fraction of channels inhibited in acute slice preparations. But the reason for this is likely different than what has been interpreted by this reviewer. To state that ASMs “only bind to the channel when the channel is depolarized, and even when the channel is depolarized for many seconds, bind very slowly to the channel.” is not consistent with prior data on ASM–channel interactions. Clarification on these points may

help the reviewer and a broader audience better understand the effects occurring here, and we appreciate being able to both address this concept here and by revising the manuscript.

First, ASMs bind activated channels and stabilize the inactivated state. It is correct that channels are more likely to enter these states when subject to voltage depolarization, but channel state is stochastic and can enter activated states near resting membrane potentials. The on-rate is fast enough that channels are blocked immediately in recordings in heterologous systems (Figure 1C). It is more likely that channel biophysical state stochasticity, along with drug concentration used herein, are likely dictating the rate at which channels accumulate block during repetitive spiking.

To address this in text, we have revised the 3rd paragraph of the introduction to better incorporate these ideas. This also helps with comments in the reviewer paragraph below.

The key experimental data on this is relegated to the Supplemental Figures. When the reader is first shown results of the effects of the inhibitor on firing in Fig 2, the presentation has been set up as if everything is perfect, and the inhibitor will be completely inhibiting either both or only one channel according to the mouse. With this presentation, it is then exceptionally striking that the cell in the middle panel of Fig 2A, labeled "Nav1.2/1.6 Inhibited" is firing action potentials very nicely even with both channels "inhibited". For a reader not already aware that there is likely only partial inhibition of each channel, the reaction will be "Huh? Shouldn't blocking both channels simply completely block excitability?". The authors do preface Fig 2 by a very brief allusion to the incomplete inhibition: "In spiking neurons, ASCs would therefore be predicted to exhibit use-dependence, progressively blocking channels in proportion to a neuron's activity rate" but this comes out of nowhere after the over-simplified picture of complete inhibition up to that point, and without any estimation of how much inhibition there is likely to be before activity, or how much induction of inhibition there is likely to be during the activity. Without this, interpreting the data in Fig 2 is basically impossible.

The key experimental data on this issue is really in Supplemental Figures 1-2 and Fig 4, and I found myself immediately ping-ponging back and forth between the Supplemental figures and the main text trying to understand what is going on with the partial inhibition. This was frustrating.

Thank you for these suggestions; they help with readability appreciably. We have re-organized the figures presented in the manuscript and emphasized details about ASCs to ensure readers can discern between near-complete blockade of channels (Figures 1-4) and activity-dependent ASC onboarding (Figures 5-7). We now present near-complete block experiments *first*, detailing the current clamp-> voltage clamp (-12 mV)-> current clamp experiments. We incorporated Supp. Fig. 1 into main Figure 1 and moved Supp. Fig. 2 into main Fig. 2.

As the reviewer notes, there are clear time-dependent effects on channel function when stepping to -12 mV, independent of GNE-4076 block. As stated previously, "We therefore focused on the 12-20 sec after voltage-clamp offset for subsequent analysis, as it is a period in which most channel-intrinsic recovery has occurred, but also a period in which we would still expect significant block from GNE-4076." We hope that reordering the manuscript as suggested and placing these results near the beginning will help with discerning between near-complete block and activity depending onboarding. By beginning with these experiments, which underscore that 100% block cannot be studied without "contamination" from native slow inactivation, we hope that the readers can better understand why data was done as presented.

In my opinion, the paper would be greatly improved by a detailed discussion of the voltage- and time-dependence of the inhibitor at the very beginning of the paper. For me, reading and digesting the paper would have been far easier if Fig 1 included a discussion of the voltage- and time-dependence of inhibition, and next Figs were then Supplemental Figs 1-2, and main Fig 4. The key questions are: how much inhibition is there before a 10-s current injection from the resting potential, and how much additional inhibition is there produced during either the 10-s bout of firing or the "on-boarding" depolarization protocol, and how long does that additional inhibition last? The most direct information on that is in the plots in Fig. 4D and Fig 4F in combination with Supplemental Fig 1, which shows that the on-boarding depolarization reduces current to about 30% of current before on-boarding. This is so central to the interpretation of all the results that I think Supp Fig 1 should be in the main paper as the first piece of data in neurons.

We originally had the nucleated patch data in supplement due to space constraints in an already large figure 1. Based on your recommendation we have moved it to the main figure. We have also changed the ordering of the paper and related figures to present data as suggested. Hopefully this better guides readers through the questions you are raising above, which are addressed in the (now reordered) figures mentioned above.

Specific:

(1) Fig.1 I can find no information on the voltage protocol used to generate the dose-response curves. In the literature characterizing sulfonamide blockers, most protocols use very unphysiological strong, long depolarization to induce inhibition, usually with equally unphysiological short hyperpolarizations to produce recovery from inactivation. One assumes something like that was used here. Obviously, the protocol needs to be explained.

We updated the methods section to better describe the voltage protocol used to generate the dose response curves. In contrast to the literature characterizing sulfonamide blockers, we used pulses that closely mimic physiological activation from -80 mV (rest) to 0 mV (depolarized) for 20 msec. GNE-4076 was perfused onto cells at increasing concentrations throughout the experiment. At each successive dose, cells were held at 0 mV to allow adequate GNE-4076 onboarding.

(2) Supp Fig1. This shows the effect of depolarization to enhance inhibition, but not how much inhibition there was before the depolarization. Presumably, there were measurements during the application of drug? How much inhibition is there before the depolarization? Why does the time only go to 20-s, when the times in Figs 4 go to 10 minutes?

Nucleated patch recordings are notoriously difficult to maintain for long durations, especially when subjecting the patch to large voltage deflections. These recordings extend to 20s recovery periods because that is the duration for which we maintained all recordings, though some exhibited rather impressive longevity and allowed for several minutes of recording thereafter. Regardless, the goal here was to assess block within the 12-20 sec recovery window we utilized in current clamp recordings from intact neurons. This was achieved.

Please note that GNE-4076 was present throughout all recordings. This was in part due to time constraints, as we could not maintain patches long enough to also perform wash-in. The degree of inhibition can be inferred by comparing peak dV/dt and threshold of cells in the absence and presence of GNE-4076. These data are presented in a new Supplemental figure 1, showing no difference in threshold or peak dV/dt.

(3) Fig. 4. Similar question here - this is a very nice and informative figure, but we see only the delta in threshold and dv/dt , but how were the initial absolute values different in the drug compared to control?

These data are presented in a new Supplemental Figure 1, showing no difference in threshold or peak dV/dt .

(4) Fig 2. As far as I can tell, we have no idea how much inhibition there is at rest, before the current injection -what is the dv/dt in the drug compared to in the control? Were there experiments in which the current injections were delivered before and after applying drug? If not, at least it would be useful to see population data on dv/dt of the first spike in control and with drug.

These data are presented in a new Supplemental Figure 1, showing no difference in threshold or peak dV/dt .

(5). Fig. 2. Do the authors have any quantitative information on how much extra inhibition would be produced at 200 nM drug using physiological waveforms of firing?

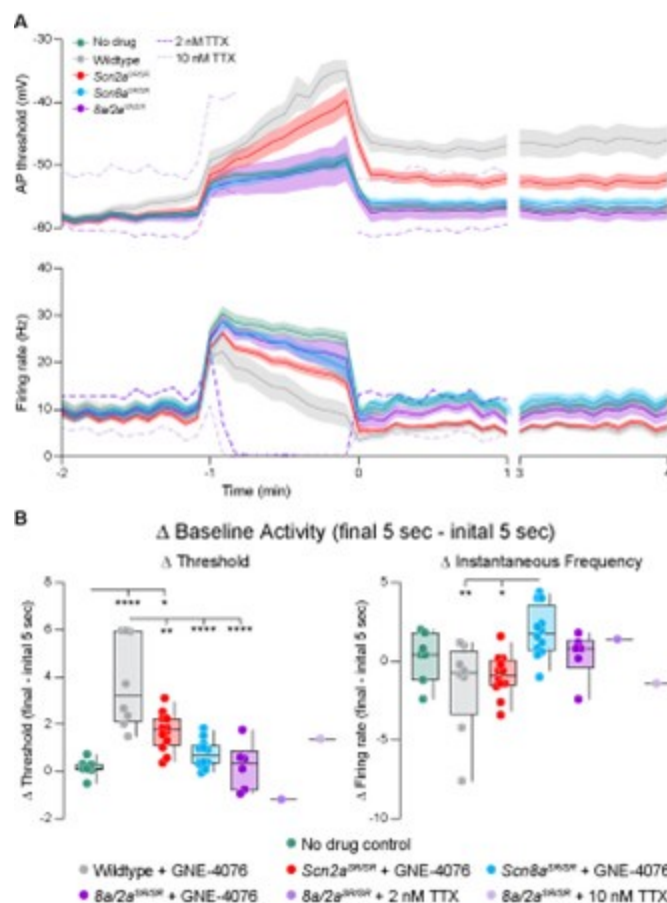
These types of analyses are part of later figures using EPSC-like waveforms to evoke spiking.

I was unconvinced that the changes in threshold and dv/dt during the firing in the drug necessarily represent time-dependent use-dependent effects of drug. Partial inhibition by TTX would probably produce greater progressive changes in spike shape and reduced ability to fire robustly.

TTX is not use-dependent, so it is a good contrast to GNE-4076. We experimented with a few cells at 2 and 10 nM TTX concentrations and found that concentrations required to mimic the block of spiking that occurs with 200 nM GNE-4076 in WT cells was associated with a marked use-independent elevation in AP threshold, with an inability to maintain ~10 Hz spiking rates with the baseline EPSC-like stimulation pattern. These effects are very different from those produced by GNE-4076, but were expected given the use-independence of TTX. We did not pursue this line of inquiry fully, so we present these data only as individual examples in the reviewer figure below:

Author response image 1.

Data from Figure 6B, D, E are replicated here with individual lines of 2 nM and 10 nM TTX shown in dashed lines. Note marked changes in threshold not observed with GNE-4076. TTX sourced from Alomone Labs.



Minor:

p. 5 and elsewhere: it seems unnecessary to give values of threshold and dv/dt to three decimal places, especially when the precision is not better than a single decimal place.

We have reduced unnecessary precision throughout.

Reviewer #3 (Recommendations for the authors):

The computational model is highly underdeveloped. Without more rigorous development the results of the computational model appear to provide little additional insight beyond that expected from the known axodendritic localizations of NaV 1.2 and 1.6. If the authors wish to use the computational results to make rigorous predictions, then this section needs to be either expanded to be more complete and promoted to a regular figure, with full details of the model, and how it was evaluated for accuracy. Alternatively, this point regarding computational insight could be de-emphasized and or removed from the paper.

Modeling:

(1) I don't see any methods describing the precise model parameters that were used.

Apologies, this is a model that we have built and tested extensively over the years (PMID: 38290518, 35417922, 34348157, 31995133, 31230762, 28256214), though there have been some small updates over these works. We have deposited this model at ModelDB and provide data there regarding model construction (access #2019342).

(2) There appears to be no robustness test to assess whether the particular results/conclusions were unduly dependent on particular model construction decisions.

We have now generated a new supplemental figure 2 that explores the robustness of these observations to changes in NaV1.2 and NaV1.6 position within the AIS and changes in relative density of NaV1.2 and NaV1.6. As shown there, the model is tolerant to all but extreme, non-physiological manipulations to these parameters.

(3) Figure S2 does not really provide convincing evidence of a biologically relevant model. Probably the model itself needs to be redesigned to better replicate the biological response and be validated by testing parameter sensitivity.

a) All of the results in S2C show that there is a huge reduction in the first action potential (black?) followed by relatively little change in subsequent spikes. This is not seen in any of the models. The progressive changes in threshold as predicted by the model for dual and NaV1.6 block are not at all evident in the results of C, except perhaps for the the very first and the very last spikes.

b) The baseline action potential in B is different than the recorded action potentials. In particular, the somatic depolarization occurs much later and over a more extended time frame than the real neuron, and the phase plot shows an actual dip in depolarization at the transition to the somatic spike, which is not representative of naturally occurring action potentials.

To address both (a) and (b), please note that in empirical experiments there are two parallel processes occurring: block by GNE-4076 and channel recovery from inactivation. In the model we can isolate the effects of block to test that parameter fully and in isolation. This is something that we could never achieve biologically. The important take home here in both cases is to observe that with NaV1.6 block there is a change in threshold, whereas with NaV1.2 block there is none.

(4) The one finding that seems to be robust is that the changes in NaV1.2 have little effect on threshold.

Yes! This is a major take-home message from both the model and the use of these knockin mice in combination with GNE-4076. In mature pyramidal cells, NaV1.6 is the major determinant of AP threshold. And to editorialize on this observation, changes in threshold are a useful metric to test if other pharmacology are truly selective for NaV1.2 over NaV1.6. We note that phrixotoxin-3, which is described as NaV1.2 specific in multiple papers, was never tested for specificity over NaV1.6 in its original description, and we find that it fails this test in our hands.

Data presentation:

(1) The phase plots in Figure 3B (left and right) appear to be visually identical, and as such don't strongly support any particular conclusion.

We changed the representative example record (specifically for Fig. 3A-B) to more directly support the conclusions.

(2) It is unclear to me what is meant by AP speed (title of Figure 3 legend). Do the authors mean propagation speed along the axon, or perhaps the rate of action potential firing?

Apologies, we are referencing dV/dt when we mention AP speed. We updated AP speed to AP velocity throughout the manuscript.

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